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LEIPZIG

FAKULTÄT FÜR PHYSIK
UND GEOWISSENSCHAFTEN

Classification and Similarity Scoring of Extended LOFAR Radio Sources via Self-Supervised Learning

Bachelor's Thesis

Submitted by:

Edgar Harutyunyan

3775171

Supervisors:

Prof. Dr. Matthias Hoeft

M.Sc. Nicolas Baron Perez

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Declaration of Authorship

I declare that this thesis is my work and that I accept responsibility for its contents. Sources, software, data, and assistance used in preparing it are identified, and material taken from other works is acknowledged. This thesis has not previously been submitted, in whole or in substantial part, for another degree or examination.

I used Large Language Model (LLM) assistants to support manuscript planning, drafting and language revision, code development and debugging, figure preparation and general aesthetics. Their outputs were critically reviewed against the code outputs, numerical checks, generated figures, and cited sources.

Place and date

Signature of Edgar Harutyunyan

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Abstract

We investigate the morphological classification of extended radio sources using self-supervised learning. The analysis covers $32\,514$ LoTSS-DR2 sources with largest angular sizes of $60''$–$1000''$. Contrastive pre-training, manual cluster curation and a 50-member classifier ensemble produce an eleven-class catalogue with morphology assignments and uncertainty estimates.

We introduce the similarity-to-radio-galaxy score S_{RG}, which combines support for radio-galaxy-like classes with ensemble reliability. The score distinguishes broad morphological resemblance from concentrated and consistent predictions, providing an additional criterion for selecting sources for visual inspection.

Comparisons with three external catalogues reveal stronger FRII than FRI correspondence in two catalogues, with similar performance in the third. Reference FRI sources frequently receive Bright-core or Structured-Bent assignments, highlighting the importance of central brightness and curvature in the learned classification. Application to ten manually isolated radio relics produces varied morphology assignments and low or transitional scores, illustrating how the framework describes diffuse emission through structures learned from radio galaxies.

Contents

Declaration of Authorship i

Acknowledgements ii

Abstract iii

List of Figures vi

List of Tables vii

1 Introduction 1

2 Background 3

2.1 LOFAR and the LoTSS survey . . . 3

2.2 Radio galaxies and the morphological taxonomy . . . 4

2.3 Diffuse radio emission in galaxy clusters . . . 6

2.4 Self-supervised representations and clustering . . . 7

2.5 Deep ensembles and predictive uncertainty . . . 9

3 Data 11

3.1 LoTSS-DR2 sample selection and preprocessing . . . 11

3.2 External reference catalogues . . . 12

3.3 Radio-relic reference sample . . . 13

4 Methodology 15

4.1 Model Pre-training . . . 15

4.2 Cluster-based class construction . . . 18

4.3 Supervised classifier and deep ensemble . . . 21

4.4 Similarity-to-radio-galaxy score S_{RG} . . . 22

4.5 External catalogue mapping and evaluation . . . 24

4.6 Application to diffuse radio emission . . . 25

5 Results 27

5.1 Internal classification and S_{RG} evaluation . . . 27

5.1.1 Ensemble classification catalogue . . . 27

5.1.2 Score distributions and operational bands . . . 29

5.2 Comparison with external morphology catalogues . . . 30

5.2.1 FRI correspondence . . . 32

5.2.2 FRII correspondence . . . 33

5.2.3 Three-catalogue consensus . . . 34

5.2.4 S_{RG} distribution across mapped-reference sources . . . 34

5.3 Application to radio relics . . . 37

6 Discussion **39**
6.1 Morphological classification and external correspondence 39
6.2 Similarity scoring and diffuse-emission applications 40
6.3 Limitations and implications for future work 40

7 Conclusion and Outlook **43**
7.1 Main conclusions 43
7.2 Future work 43
7.3 Data and software availability 43

A GreyResNet18 encoder and projection head **45**

B Raw PCA–HDBSCAN cluster overview **46**

C Examples of external-label disagreement **49**

D Supplementary evaluation **52**

References **53**

List of Figures

1 LOFAR station network and Tautenburg station . . . 3
2 Examples of radio-galaxy morphologies . . . 5
3 Examples of radio haloes, relics and mini-haloes . . . 6
4 Radio and X-ray emission in the Sausage cluster . . . 7
5 Angular-size and peak-flux-density distributions . . . 12
6 SimCLR pre-training with random structural views . . . 16
7 HDBSCAN cluster membership distribution . . . 17
8 Examples of six raw HDBSCAN clusters . . . 17
9 PCA projection of curated classes and unassigned sources . . . 19
10 Examples of the eleven internal morphology classes . . . 20
11 Deep-ensemble inference and similarity scoring . . . 22
12 Highest- and lowest-confidence examples by class . . . 28
13 Full-catalogue score distributions . . . 29
14 Similarity-score distributions by predicted class . . . 30
15 Class correspondence with external catalogues . . . 31
16 Predicted classes of missed reference FRI sources . . . 32
17 Ensemble confidence in the FRI comparisons . . . 32
18 Predicted classes of missed reference FRII sources . . . 33
19 Ensemble confidence in the FRII comparisons . . . 33
20 Predicted classes of three-catalogue consensus sources . . . 34
21 Score distributions for mapped-reference sources . . . 35
22 Examples of agreement with external catalogues . . . 36
23 Masked radio-relic inputs and predicted classes . . . 37
24 Similarity scores for selected radio relics . . . 38
25 GreyResNet18 encoder and projection head . . . 45
26 HDBSCAN overview: noise and clusters 0–28 . . . 46
27 HDBSCAN overview: clusters 29–58 . . . 47
28 External disagreements: Amorphous, ACJ, Bright-core and FRI . . . 49
29 External disagreements: Core-Displaced, FRII, Hybrid and Large FRII . . . 50
30 External disagreements: Spirals, Structured-Bent and Unresolved . . . 51

List of Tables

1 Example ensemble probabilities and agreement . . . 9
2 Individual reference-catalogue selections . . . 12
3 Combined reference-catalogue selections . . . 13
4 Selected radio relics and host clusters . . . 13
5 Labelled subsample and SI-SSIM retention fractions . . . 19
6 Correlations between ensemble reliability measures . . . 23
7 Mapping between external and internal morphology classes . . . 24
8 Class composition, confidence and agreement . . . 27
9 Catalogue membership by score band . . . 29
10 External correspondence excluding curated sources . . . 30
11 FRI agreement with external catalogues . . . 32
12 FRII agreement with external catalogues . . . 33
13 Agreement with three-catalogue consensus labels . . . 34
14 Group assignments for fine-class agreements and disagreements . . . 35
15 Ensemble predictions and scores for selected relics . . . 37
16 Raw-cluster consolidation into morphology classes . . . 48
17 Test-set classification counts . . . 52
18 Angular-size sensitivity of external correspondence . . . 52

Chapter 1. Introduction

Modern wide-area radio surveys such as the LOFAR Two-metre Sky Survey (LoTSS) are producing millions of radio sources with diverse morphologies. Visual classification of objects at this scale is time-consuming and does not provide a scalable approach to the analysis of source morphology (Baron Perez et al., 2025).

Radio morphology reveals the activity of the central active galactic nucleus (AGN), the propagation of its jets, and their interaction with the surrounding medium, thereby providing information about nuclear activity, the central engine, and the source environment (Baron Perez et al., 2025; Horton et al., 2025). The observed structures are diverse, and apparently similar extended emission can arise from radio galaxies or from physically distinct diffuse cluster sources (van Weeren et al., 2019). A scalable analysis must therefore capture morphological diversity while recognising that visual similarity alone does not determine physical source identity.

Supervised machine learning (ML) approaches require labelled training samples, but available labels cover only part of this diversity and may follow incompatible classification schemes (Baron Perez et al., 2025). Self-supervised learning (SSL) instead derives representations from unlabelled images and can organise radio sources according to morphological structure. Baron Perez et al. (2025) applied this approach to smaller resolved sources in the second data release (LoTSS-DR2) with largest angular sizes (LAS) in the range $30'' < \mathrm{LAS} < 60''$, leaving the larger resolved-source regime unexplored.

In this thesis, we address that gap by extending the morphology-learning framework of Baron Perez et al. (2025) to $32\,514$ LoTSS-DR2 sources with $60'' \leq \mathrm{LAS} \leq 1000''$. Its primary result is an 11-class catalogue that records an ensemble morphology assignment and uncertainty summaries for every selected source. The central methodological contribution of the thesis is the *similarity-to-radio-galaxy* score S_{RG}, which is proposed as an uncertainty-aware way to prioritise sources whose image morphology differs from the radio-galaxy-like classes learned by the model.

The resulting morphology classes are evaluated through comparison with three independent catalogues. The analysis describes the behaviour of S_{RG}. The relic application illustrates this behaviour on selected radio relics but is not interpreted as evidence of physical source classification.

We ask which morphology groups emerge in this sample, how their predictions correspond to internal and external labels, and how the proposed score responds to uncertainty and diffuse emission.

The encoder, contrastive framework, structural-view augmentation and classifier pipeline come from Baron Perez et al. (2025). Our contributions are the adapted selection and training configuration, curated taxonomy, similarity score, catalogue comparisons and masked-relic study, supported by the polygon interface and analysis tools.

The remainder of the document is organised as follows. Chapter 2 reviews the LOFAR survey context, radio-galaxy taxonomy, contrastive SSL, and ensemble uncertainty quantification. Chapter 3 describes the LoTSS-DR2 source selection and cutout preprocessing. Chapter 4 details the pre-training, cluster-based label construction, supervised ensemble, and definition of S_{RG}. Chapter 5 presents the catalogue and score distributions, alignment with three external catalogues, and the provisional relic test. Chapters 6 and 7 interpret the supported results, delimit the score's use, and give an outlook.

Chapter 2. Background

The interpretation of radio-source morphology requires both an astronomical context and a description of the statistical methods used to analyse the images. This chapter introduces LOFAR and the LoTSS survey, radio galaxies and diffuse cluster emission, followed by self-supervised learning, clustering and ensemble uncertainty.

2.1 LOFAR and the LoTSS survey

The LOw-Frequency ARray (LOFAR) is currently the largest radio telescope operating at the lowest frequencies observable from Earth.[1] It is a pan-European aperture-synthesis radio interferometer (Fig. 1) operating between 10 and $240\,\mathrm{MHz}$ (van Haarlem et al., 2013). Compared with contemporary centimetre-wavelength facilities (such as the VLA or MeerKAT), LOFAR offers exceptional surface-brightness sensitivity to extended, low-surface-brightness synchrotron emission at metre wavelengths. This property makes it particularly well suited to the study of diffuse Galactic and intracluster structures, while still delivering arcsecond-resolution images of the more compact emission associated with active galactic nuclei.

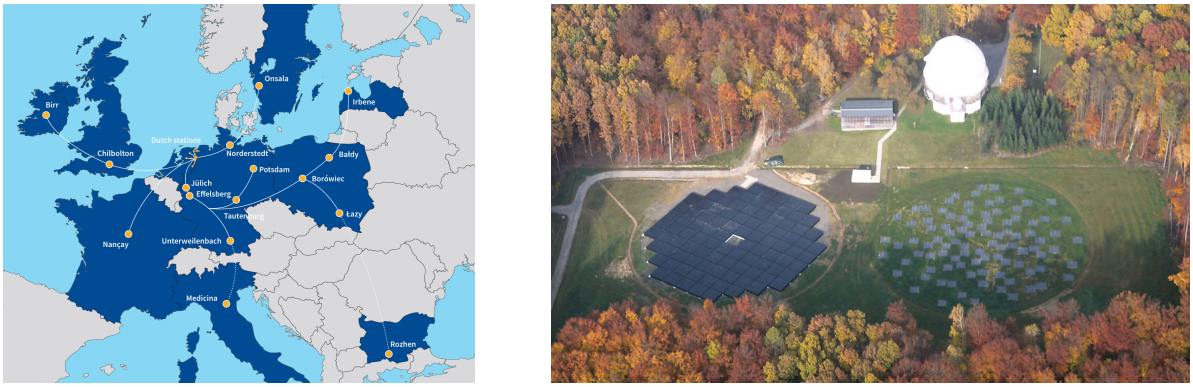


Figure 1: The Dutch and international LOFAR station network map (left; map credit: ASTRON) and the LOFAR station at Tautenburg, Germany (right; photograph by Michael Pluto).

The LOFAR Two-metre Sky Survey (LoTSS) is the principal northern-hemisphere extragalactic survey conducted with this instrument. LoTSS observes with the High-Band Antenna from 120 to $168\,\mathrm{MHz}$, with a central frequency of $144\,\mathrm{MHz}$. Its second data release, LoTSS-DR2, covers $5{,}634\,\mathrm{deg}^2$ in two contiguous regions of the northern sky at an angular resolution of $6''$, with a median rms noise of $83\,\mu\mathrm{Jy\,beam}^{-1}$, and yields a total catalogue of $4{,}396{,}228$ radio sources (Shimwell et al., 2022). At this resolution, a source with $\mathrm{LAS} = 60''$ spans approximately ten synthesised beam widths, permitting resolved morphological information to be recovered, although the visibility of individual cores, lobes, hotspots, and bent tails also depends on their surface brightness and structure.

For radio-galaxy studies, catalogue components must be associated into physical radio sources and linked, where possible, to optical or infrared host galaxies. For LoTSS-DR2, these associations are provided by a value-added catalogue combining likelihood-ratio cross-matching against optical and infrared imaging, the citizen-science project Radio Galaxy Zoo: LOFAR, and direct visual inspection by domain experts (Hardcastle et al., 2023). Within the area covered by suitable optical imaging, the

[1] `https://www.astron.nl/telescopes/lofar/`

catalogue assigns an optical or infrared identification to $85\,\%$ of LoTSS-DR2 sources and a spectroscopic or photometric redshift to $58\,\%$. It also supplies source associations and size information needed to construct a catalogue of resolved radio galaxies.

The third data release, LoTSS-DR3, substantially extends this coverage: it contains $12{,}950\,\mathrm{h}$ of observations over $19{,}035\,\mathrm{deg}^2$, corresponding to $88\,\%$ of the northern sky, and catalogues $13{,}667{,}877$ radio sources at $6''$ resolution, increasing to $9''$ below declination 10° (Shimwell, T. W. et al., 2026). A dedicated catalogue of galaxy clusters with diffuse radio emission has also been compiled from the DR3 imaging (Stuardi et al., 2026). The catalogue-scale sample in this thesis remains based on DR2 because it was selected using the host associations and angular-size information in the DR2 value-added catalogue. We describe the precise data products and selection criteria in Chapter 3.

2.2 Radio galaxies and the morphological taxonomy

Radio galaxies are radio-loud active galactic nuclei whose central engines launch collimated plasma jets from the vicinity of an accreting supermassive black hole. Synchrotron emission traces a compact radio core, the jets, and the lobes or plumes inflated where the jets deposit energy into the surrounding medium. Shocks near the ends of powerful jets can produce compact hotspots. Hence, the observed morphology encodes both the transport of energy by the jets and their interaction with the gaseous environment across galactic and cluster scales (Hardcastle & Croston, 2020). The most widely used morphological division is the Fanaroff–Riley (FR) classification. Fanaroff & Riley (1974) defined the ratio R_{FR} of the separation of the highest-surface-brightness regions on opposite sides of the host, excluding the compact central core, to the total source extent. FR I sources have $R_{\mathrm{FR}} < 0.5$ and are centre-brightened, whereas FR II sources have $R_{\mathrm{FR}} > 0.5$ and are edge-brightened (top-left and top-middle sources in Fig. 2). In the original 3CR sample, the transition occurred near a 178-MHz radio power of $P_{178} \simeq 2 \times 10^{25}\,\mathrm{W\,Hz^{-1}\,sr^{-1}}$ under the cosmology adopted in that work. This was an empirical correlation, not a strict physical boundary. Lower-power jets are typically more susceptible to disruption and deceleration into diffuse plumes, whereas more powerful jets can remain collimated until they terminate in hotspots. Nevertheless, jet power, host-galaxy properties, and environment all influence the resulting morphology (Hardcastle & Croston, 2020).

Clearly, the radio-galaxy population is not exhausted by this binary scheme. FR 0 sources are compact, core-dominated radio galaxies with weak or absent extended emission (Baldi et al., 2015). Bent-tail sources are distinguished mostly by geometry rather than by a unique formation mechanism. Narrow-angle-tail (NAT) sources have closely swept-back jets or tails and are commonly associated with ram pressure produced by relative motion between the galaxy and the intracluster medium (ICM) (top-right source in Fig. 2). Wide-angle-tail (WAT) sources have a broader opening angle, often with initially collimated jets that flare into bent plumes around a cluster-dominant galaxy (bottom-left in Fig. 2). Relative galaxy–ICM motion associated with cluster mergers can contribute to their bending (Sakelliou & Merrifield, 2000). Head-tail (HT) is a broader descriptive label for emission trailing away from a bright head and does not require a single resolved tail (bottom-middle in Fig. 2). These forms encode interactions between the radio jets, the motion of the host, and the surrounding gas. Relaxed double (RD) sources are symmetric relaxed structures, with no visible jets and less interaction with a dense environment (bottom-right in Fig. 2). The absence of compact cores, hotspots and jets suggests that these RDs represent the late evolutionary stage of radio galaxies (Chilufya et al., 2025).

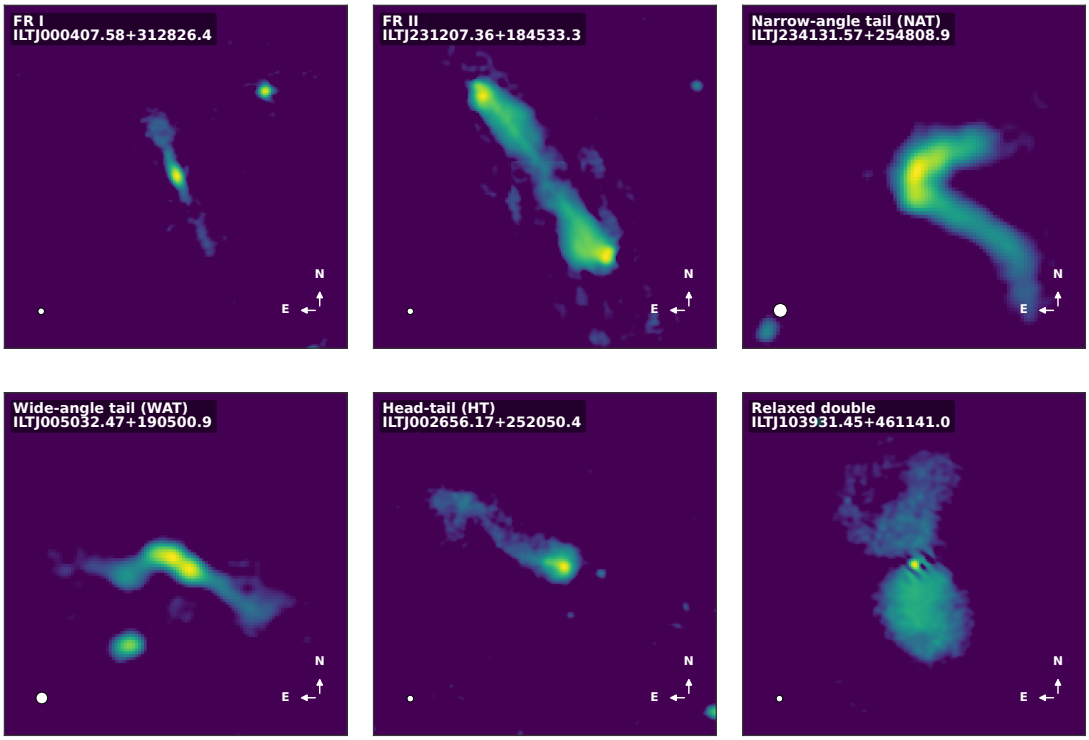


Figure 2: Representative host-centred LoTSS-DR2 144-MHz images of the radio-galaxy morphologies classified by Chilufya et al. (2025). Each field spans $1.5 \times \mathrm{LAS}$; the synthesised beam is shown at lower left, with north up and east left.

LoTSS has demonstrated that these morphological categories overlap strongly in their physical properties. Using $5{,}805$ extended radio-loud AGN from LoTSS-DR1, Mingo et al. (2019) found significant overlap between the FR I and FR II luminosity distributions and a population of FR IIs extending three orders of magnitude below the canonical FR break at $L_{150} \sim 10^{26}\,\mathrm{W\,Hz^{-1}}$. In LoTSS-DR2, Clews et al. (2025) classified $3{,}590$ FR IIs and $2{,}354$ FR Is at $z \leq 0.8$. Near the FR divide, FR I morphology is more frequent in more massive hosts, but no robust correlation between the break luminosity and stellar mass remains after selection effects are considered. This dependence on multiple source and host properties reinforces that FR morphology is not determined by a single luminosity division. Chilufya et al. (2025) studied $2{,}828$ extended radio-loud AGN with $\mathrm{LAS} > 60''$ and found that almost all FR Is in their spectroscopic sample are low-excitation radio galaxies, whereas high-excitation objects occur predominantly among luminous FR IIs; nevertheless, no single radio or host property fully determines accretion mode. Finally, Horton et al. (2025) catalogued $9{,}985$ large, bright LoTSS-DR2 sources using broad morphologies and finer structural attributes, finding at least one possible jet-precession indicator in $28\,\%$ of the sample. Together, these results motivate a morphology taxonomy richer than a strict FR dichotomy.

Radio galaxies can also be distinguished by their physical scale and activity history. For instance, giant radio galaxies have projected sizes above approximately $0.7\,\mathrm{Mpc}$; Dabhade et al. (2020) identified 239 in LoTSS-DR1, including seven larger than $2\,\mathrm{Mpc}$. Remnant and restarted radio galaxies instead preserve diffuse plasma from earlier episodes of jet activity, sometimes alongside newly formed jets or lobes. These properties can coexist with the FR and bent-tail labels and further broaden the range of radio-galaxy appearances.

Large morphology catalogues also differ in how their labels are constructed. Clews et al. (2025) used a semi-automated method based on radio-source ridgelines, whereas Horton et al. (2025) assigned detailed, multi-attribute labels through visual inspection of LoTSS images. Chilufya et al. (2025) used visual classification supported by higher-resolution VLASS observations and multiple independent classifiers. Differences in source selection, available image information, and classification specificity mean that these catalogues provide distinct reference schemes instead of establishing ground-truth labels.

2.3 Diffuse radio emission in galaxy clusters

Galaxy clusters contain a hot, magnetised intracluster medium (ICM) in which relativistic electrons produce extended synchrotron emission. This diffuse emission is not associated one-to-one with an optical galaxy and is conventionally divided into radio haloes, mini-haloes, and radio relics (van Weeren et al., 2019) (Figure 3). These categories describe different locations and morphologies within a galaxy cluster. Interestingly, they do not imply that active galaxies are irrelevant to this emission: fossil plasma deposited by earlier AGN activity may provide seed particles that are subsequently re-energised in the ICM.

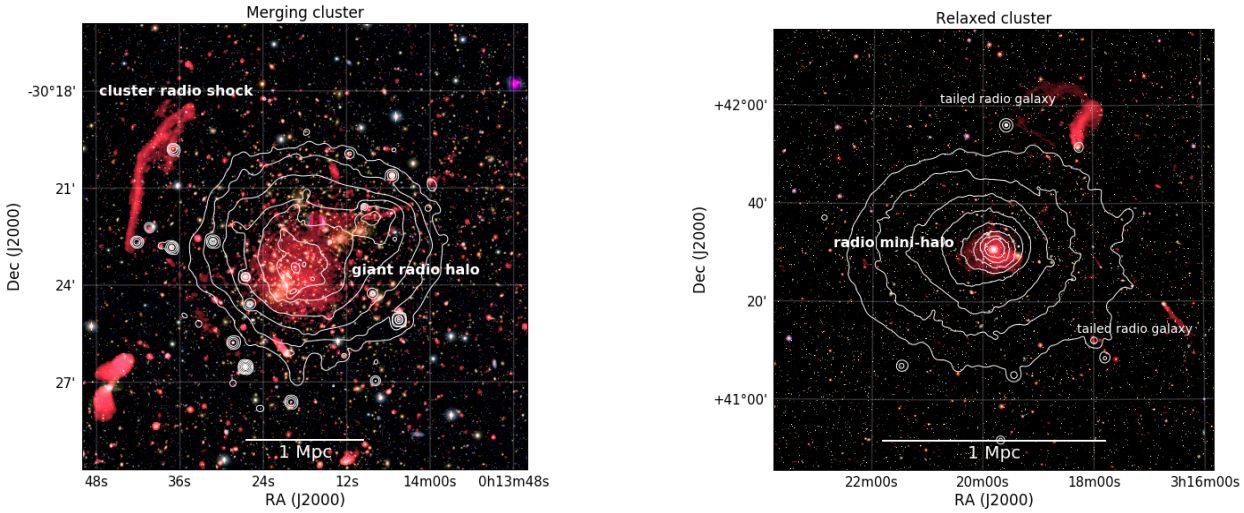


Figure 3: Radio halo and relic emission in Abell 2744 (left) and the Perseus mini-halo (right). Reproduced from van Weeren et al. (2019) under CC BY 4.0.

Radio relics are often elongated, arc-like sources found mainly in the periphery of merging clusters and are frequently polarised at gigahertz frequencies. Their locations and spectral properties associate them with merger-driven shock waves, where electrons may be freshly accelerated or re-accelerated from a fossil population. Revived fossil or phoenix plasma is a related but distinct diffuse-emission category; here, relic refers to the selected elongated cluster sources (van Weeren et al., 2019). The northern "Sausage" relic in CIZA J$2242.8{+}5301$ is a well-studied example (Fig. 4). It forms a thin, nearly $2\,\mathrm{Mpc}$ arc with aligned magnetic fields and a radio spectrum that steepens downstream from the outer edge (van Weeren et al., 2010).

Radio haloes instead occupy the central regions of merging clusters, extend over scales of order a megaparsec, and broadly follow the distribution of the thermal ICM. Turbulent re-acceleration following a merger is a leading explanation for their distributed emission. Mini-haloes are smaller, typically spanning about 100–$500\,\mathrm{kpc}$ around the brightest cluster galaxy in relaxed cool-core clusters.

Sloshing-induced turbulence and re-acceleration of seed particles supplied by the central radio galaxy are among the proposed mechanisms. Haloes and mini-haloes can contain substantial substructure, but unlike relics their emission is not localised to an identifiable peripheral shock (van Weeren et al., 2019).

Radio relics have a different physical origin from radio galaxies, yet their diffuse structures may resemble parts of radio-galaxy emission. This motivates examining whether a model trained to distinguish radio-galaxy morphologies assigns similar descriptions to relics. We explore this question using selected, manually masked relic images described in Section 3.3. Nevertheless, this test does not establish whether the model can recognise unfamiliar sources in general.

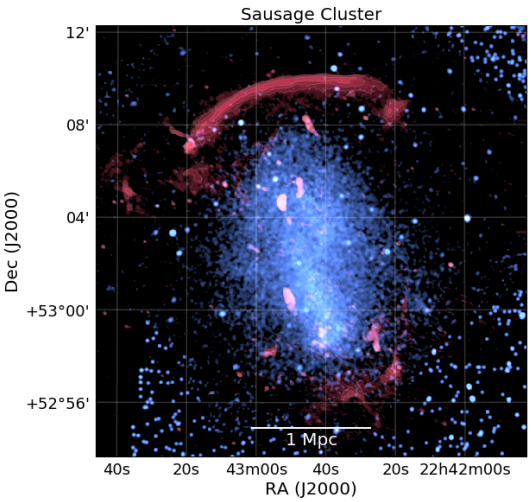


Figure 4: The Sausage cluster: 610-MHz radio (red) and X-ray (blue) emission. Reproduced from van Weeren et al. (2019) under CC BY 4.0.

2.4 Self-supervised representations and clustering

Machine learning (ML) uses models fitted to data to recognise patterns and make predictions. In astronomy, it provides tools for analysing large survey datasets whose scale and diversity make detailed manual inspection difficult. Deep learning is a subset of ML in which neural networks with multiple processing layers organised within a particular architecture learn successive representations of the input. For image inputs, these representations can capture spatial structure directly from pixel intensities. Applications in radio astronomy include source detection, morphological classification, and searches for similar or unusual objects (Slijepcevic et al., 2024; Riggi et al., 2024). However, the available images greatly outnumber those with reliable expert labels, motivating methods that can learn useful representations from unlabelled observations (Slijepcevic et al., 2024).

Supervised learning trains a model from images paired with class labels, and unsupervised learning searches for structure without such labels. Self-supervised learning also uses unlabelled data, but constructs a training objective from the data themselves. In this thesis, self-supervised pre-training first teaches a network to encode radio-source structure. The pre-trained network is then fine-tuned on a smaller labelled sample for morphological classification (Baron Perez et al., 2025).

An encoder f_θ maps an input image x to a finite-dimensional representation $h = f_\theta(x)$, also called an embedding. Distances in the embedding space express which image features the encoder regards as similar. A morphology-aware embedding should place structurally similar radio sources close together even when their orientation or brightness differs. Subsequently, this representation space can be clustered to construct morphological groups and used as the input to a classifier (Baron Perez et al., 2025; Slijepcevic et al., 2024).

The framework adopted by Baron Perez et al. (2025) and followed here is *A Simple Framework for Contrastive Learning of Visual Representations* (SimCLR; Chen et al., 2020). Two independently augmented views of each image form a positive pair. Views originating from other images in the same minibatch act as negative examples. These comparisons provide a supervisory signal without requiring morphology labels. The augmentations are assumed to preserve the underlying source morphology while

changing nuisance properties such as orientation or brightness. Bringing the resulting positive views together therefore teaches the encoder invariance to those transformations. A minibatch is the group of images processed together in one optimisation step. An encoder f_θ and a projection head g_ϕ are trained jointly so that projected positive-pair embeddings are similar and negative-pair embeddings are separated. For $2N$ views generated from N images, the contribution of a positive pair (i, j) to the normalised temperature-scaled cross-entropy (NT-Xent) loss is

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k\neq i]} \exp(\mathrm{sim}(z_i, z_k)/\tau)}, \tag{2.1}$$

where $z = g_\phi(f_\theta(x))$, $\mathrm{sim}(u, v) = u \cdot v/(\|u\|\|v\|)$ is cosine similarity, and τ controls how strongly the loss distinguishes similar from dissimilar embeddings. The indicator excludes the anchor from the denominator. After pre-training, the projection head is discarded and the encoder is retained for downstream use.

For an anchor view i, the numerator measures its similarity to the corresponding positive view j. The denominator compares that positive similarity with the similarities between i and every other view in the minibatch, excluding only i itself. Minimising the loss therefore increases the positive pair's similarity relative to the competing views. The temperature τ controls how strongly differences between similarities are emphasised; each of the two views is used in turn as the anchor, and the resulting contributions are averaged.

Standard SimCLR has a limitation for radio-morphology applications: views originating from other images in the minibatch are treated as negative examples. Distinct sources with nearly identical morphology may therefore be pushed apart in the representation space. The random structural view (RSV) introduced by Baron Perez et al. (2025) addresses this limitation by allowing one augmented view to be replaced by a view of a structurally similar source identified using semi-invariant structural similarity. Thus, positive pairs can connect distinct sources with similar morphology, encouraging the representation space to encode cross-source structural similarity. The transformations and selection parameters used in this thesis are specified in Sec. 4.1.

After pre-training, the learned representations must be organised into candidate morphological groups. Principal component analysis (PCA) projects the high-dimensional representations onto orthogonal axes ordered by the variance they explain (Jolliffe & Cadima, 2016). This linear construction is mathematically transparent: every principal component is a linear combination of the encoder features, its contribution is quantified by its explained variance, and the reduced dimensionality can be selected through an explicit retained-variance criterion. Hierarchical density-based spatial clustering of applications with noise (HDBSCAN) is applied here to the retained multidimensional PCA representation. Unlike methods that require a predetermined number of approximately spherical clusters, HDBSCAN identifies dense regions with potentially different shapes and densities and can leave isolated sources unassigned as noise (Campello et al., 2013; McInnes et al., 2017). These properties are useful for an uncurated radio-source sample containing rare morphologies, artefacts, and ambiguous sources. For visualisation, the first three principal components can be plotted and coloured by the HDBSCAN assignments. Visual inspection of the source images within each resulting cluster then assesses morphological coherence and supports consolidation into a labelled subsample. The adopted PCA and HDBSCAN configuration is specified in Sec. 4.1.

2.5 Deep ensembles and predictive uncertainty

The output probabilities of a single neural network by themselves can be overconfident. They do not quantify epistemic uncertainty due to training data limitations or model choice. To increase the reliability of the probability estimates for the class predictions, Baron Perez et al. (2025) adopted a deep ensemble. Lakshminarayanan et al. (2017) proposed that training an ensemble of randomly initialised networks (M classifiers with the same architecture) can yield better-calibrated probabilities and stronger out-of-distribution detection, while training and inference costs scale approximately with ensemble size.

Let $\mathbf{p}_m(x) = (p_m^c(x))_{c\in\mathcal{K}}$ denote the class-probability vector predicted for source image x by ensemble member $m \in \{1, \ldots, M\}$, where $\mathcal{K}$ is the class set with cardinality $K = |\mathcal{K}|$. The ensemble mean vector $\bar{\mathbf{p}}(x)$ and ensemble agreement are defined as

$$\bar{p}^c(x) \;=\; \frac{1}{M}\sum_{m=1}^{M} p_m^c(x), \qquad A(x) \;=\; \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\left[\arg\max_{c\in\mathcal{K}} p_m^c(x) = \hat{y}\right], \tag{2.2}$$

where $A(x) \in [0, 1]$ measures the fraction of ensemble members whose top-predicted class coincides with the ensemble-mean decision $\hat{y} = \arg\max_{c\in\mathcal{K}} \bar{p}^c(x)$. The total predictive uncertainty of the ensemble is captured by the entropy of the mean distribution,

$$H[\bar{\mathbf{p}}](x) \;=\; -\sum_{c\in\mathcal{K}} \bar{p}^c(x)\, \log \bar{p}^c(x), \tag{2.3}$$

which is maximised at $\log K$ for a uniform distribution over the K classes.

member / class	0	1	2	3	4	5	6	7	8	9	10	row sum
$m = 1$	0.02	0.03	0.03	0.05	0.48	0.04	0.04	0.05	0.02	0.17	0.07	1.00
$m = 2$	0.03	0.02	0.04	0.04	0.38	0.05	0.04	0.05	0.03	0.24	0.08	1.00
$m = 3$	0.04	0.03	0.03	0.04	0.30	0.06	0.05	0.04	0.04	0.27	0.10	1.00
$m = 4$	0.03	0.04	0.02	0.03	0.20	0.07	0.05	0.04	0.05	0.32	0.15	1.00
mean $\bar{p}^c$	**0.03**	**0.03**	**0.03**	**0.04**	**0.34**	**0.055**	**0.045**	**0.045**	**0.035**	**0.25**	**0.1**	**1.00**

Table 1: Illustrative class-probability vectors for four ensemble members and their component-wise mean. Each member's predicted class is the class with the largest probability in its row, whereas the ensemble decision $\hat{y}$ is the class with the largest mean probability. Here, $\hat{y} = 4$ and three of the four members also predict class 4, giving $A = 3/4$. Each row sums to one.

The total uncertainty $H[\bar{\mathbf{p}}]$ can be decomposed algebraically into terms commonly associated with the data (aleatoric) and model (epistemic) components. The model-disagreement term, commonly interpreted as epistemic uncertainty, is the mutual information (MI) between the class prediction and the ensemble index m when members are treated as uniformly sampled models. It can be expressed in closed form as the difference between the entropy of the mean and the mean of the per-member entropies (Houlsby et al., 2011; Depeweg et al., 2018),

$$\mathrm{MI}(x) \;=\; H[\bar{\mathbf{p}}](x) \;-\; \frac{1}{M}\sum_{m=1}^{M} H[\mathbf{p}_m](x). \tag{2.4}$$

All ensemble members start from the same pre-trained encoder weights and biases, which are subsequently updated during fine-tuning. The resulting independently fine-tuned members are not exact samples from a posterior distribution. Therefore, $\mathrm{MI}(x)$ is interpreted here as an empirical measure of model disagreement, not as an exact Bayesian separation of aleatoric and epistemic uncertainty.

The mutual information is bounded above by $\log K$, as is H. Compared to the agreement rate A, MI captures disagreement between ensemble members across the full predicted distribution. It can therefore respond to differences that leave the top class unchanged, but it is not universally more sensitive than A. Henceforth, the quantities (A, H, MI) are used as complementary uncertainty-related inputs to define the reliability factor $R(x)$ (see Sec. 4.4).

Chapter 3. Data

Three datasets are used throughout this work: the extended LoTSS-DR2 source sample, external morphology catalogues, and selected radio-relic cutouts. This chapter describes their selection and preparation, distinguishing the images used for model training and catalogue inference from the reference data used for comparison and the diffuse-emission application.

3.1 LoTSS-DR2 sample selection and preprocessing

The main catalogue dataset consists of the two LoTSS-DR2 products introduced in Sec. 2.1:

- the $6''$-resolution Stokes I mosaics provide the image data, with their $1.5''$ pixel sampling used to construct the cutouts (Shimwell et al., 2022); and
- the value-added catalogue of Hardcastle et al. (2023) supplies the source identifier, resolved-source flag, optical or infrared host coordinates, largest angular size (LAS), peak flux density, and mosaic identifier.

These quantities define the sample and extraction geometry. Note that no morphology label from an external catalogue enters this stage.

We started from the *resolved* sources of the value-added catalogue and applied the following filters:

- an angular size $60'' \leq \mathrm{LAS} \leq 1000''$;
- an optical or infrared host position with both coordinates available; and
- a peak flux density of at least $0.75\,\mathrm{mJy\,beam^{-1}}$.

The host coordinates define the centre of each cutout. Unlike Baron Perez et al. (2025), we did not impose an active-pixel selection because it would substantially reduce the sample and could preferentially remove sources whose emission is distributed across diffuse, low-surface-brightness structures. The resulting dataset contains $32\,514$ sources and is used for representation learning and catalogue inference.

The angular-size window selects the extended-source regime motivated in Sec. 2.1 and complements the $30'' < \mathrm{LAS} < 60''$ interval studied by Baron Perez et al. (2025). The upper bound of $1000''$ limits both the downsampling required for the fixed model input size and the amount of unrelated emission in the frame. Centring each cutout on the host position rather than the radio centroid avoids systematically anchoring asymmetric sources at their brightest component (Baron Perez et al., 2025).

The selected population is concentrated towards the lower ends of both the angular-size and peak-flux ranges (Fig. 5).

Each selected source is converted into a model input by the preprocessing pipeline of Baron Perez et al. (2025):

1. A square cutout of side $1.5 \times \mathrm{LAS}$ is extracted from the corresponding mosaic and centred on the host position.
2. Pixel values below $v_{\min} = 3\sigma_{\mathrm{img}}$ are set to $v_{\min}$, where σ_{img} is the sigma-clipped standard deviation of the cutout, estimated with a 3σ rejection threshold and at most ten iterations. This suppresses background fluctuations and low-intensity artefacts.

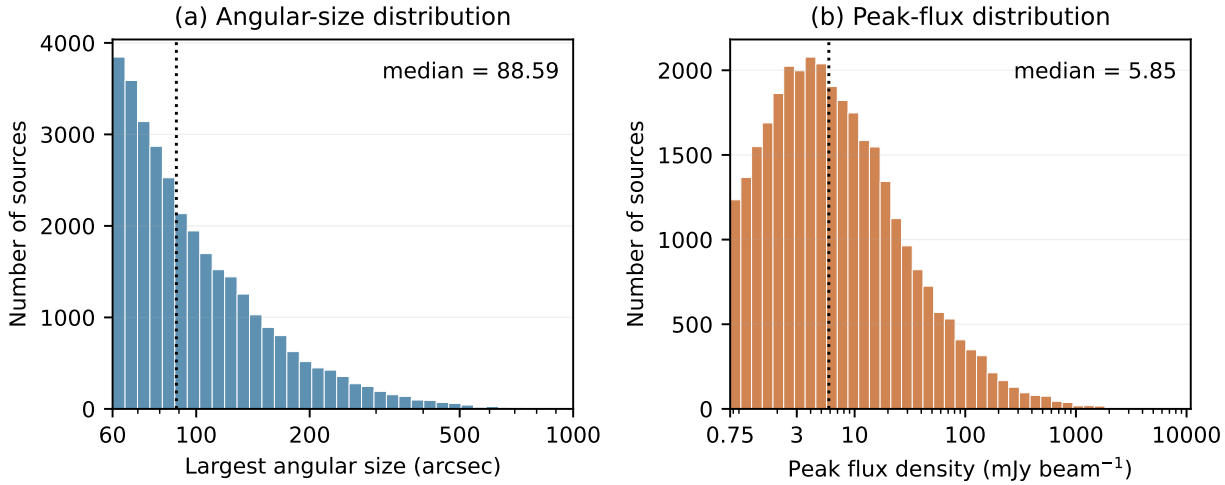


Figure 5: Largest-angular-size and peak-flux-density distributions of the 32 514 sources in the processed catalogue sample. Both horizontal axes are logarithmic; dotted lines mark the sample medians.

3. The clipped image is then resampled to 128×128 pixels using scikit-image's default first-order interpolation for floating-point arrays, min–max normalised to $[0, 1]$, and stored as an 8-bit single-channel greyscale image.

Because the cutout side scales with the LAS, sources of very different angular size appear at a comparable scale in the frame, so that the network input is dominated by morphology.

The model receives linearly min–max-normalised intensities. For visual inspection, the representative-source galleries use a symmetric-logarithmic intensity display to reveal faint structure. Only the archived raw-cluster panels retain their original display scaling.

3.2 External reference catalogues

We used three independently constructed morphological catalogues (Table 2) of LoTSS-DR2 sources exclusively for comparison (Sec. 5.2).

Catalogue	Description	Full catalogue	Matched	Mapped	Mapped, curated excluded
Clews et al. (2025)	Semi-automated ridgeline analysis covering classical radio-galaxy morphologies.	20 172	11 062	8 402	7 757
Horton et al. (2025)	Visual inspection with multi-attribute labels, including hybrid and precession-related signatures.	9 985	9 287	9 070	8 352
Chilufya et al. (2025)	Classification of extended radio-loud AGN selected with $\mathrm{LAS} > 60''$.	2 828	2 788	2 788	2 590

Table 2: Reference catalogues and source counts after identifier matching, morphology mapping and exclusion of internally curated sources.

Firstly, we cross-compared each catalogue with our primary dataset by matching their `Source_Name`

identifiers. These are flagged as **Matched** in Table 2. We then translate the reference labels into our morphology classes using the rules in Sec. 4.5, retaining sources with usable labels as the **Mapped** sample. Reference labels come from Clews's `Morph`, Horton's boolean flags and Chilufya's `Final_Morph` (`radio_optical_class-v1`). In Sec. 5.2, we report F_1 first for the **Mapped, curated excluded** samples, which remove overlap with supervised training, and then for the full **Mapped** samples.

Besides individual usage of these three catalogues, we used sources with identical mapped labels in all three catalogues. Table 3 details each sample and its purpose.

Sample	Sources	Definition
At least one mapped reference	16 177	Unique study sources with a usable label in at least one catalogue; used for score distributions.
Common-source sample	404	Study sources present in all three catalogues, regardless of label availability.
Common mapped sample	311	Common sources with usable mapped labels in all three catalogues.
Consensus sample	149	Common mapped sources assigned the same morphology by all three catalogues.
Consensus excluding curated sources	139	Consensus sources remaining after removing the internally curated sample.

Table 3: Definitions and source counts for combined reference-catalogue selections.

3.3 Radio-relic reference sample

For the diffuse-emission application in Sec. 4.6, we select ten radio relics from a yet-unpublished catalogue compiled by The Tautenburg Radio Relic Collection team under the supervision of Prof. Dr. Matthias Hoeft. Table 4 lists their relic-centred coordinates and largest angular sizes (LAS), which we use directly to construct the cutouts from LoTSS-DR3. Their host clusters and the reference positions are also provided for identification.

	Relic position				Cluster position	
Relic	RA	Dec	LAS	Host cluster	RA	Dec
P113 N	4.9363	25.3543	228	PSZ2 G113.91-37.01	4.9129	25.2908
Toothbrush D	90.9042	42.1708	66	RX J0603.3+4214	90.825	42.2333
Toothbrush B	90.8333	42.3083	474	RX J0603.3+4214	90.825	42.2333
MS1752 SW	267.9745	44.6325	192	MACS J1752.0+4440	268.009	44.6773
MS1752 NE	268.0829	44.6953	270	MACS J1752.0+4440	268.009	44.6773
Zw2341 RS	355.95	0.23	216	ZwCl 2341.1+0000	355.919	0.3053
P096p N	284.1583	66.4417	204	PSZ1 G096.89+24.17	284.16	66.3861
P096p S	284.1875	66.3292	330	PSZ1 G096.89+24.17	284.16	66.3861
P116 N	8.0645	18.1766	103.8	PSZ2 G116.50-44.47	8.0342	18.1069
A2256 R1	256	78.7667	600	Abell 2256	255.9064	78.623

Table 4: Selected relics and their host clusters. Coordinates are in degrees and LAS in arcseconds.

We note that the original FITS headers give a circular $9''$ beam for Zw2341 RS and $6''$ beams for the other nine selected cutouts. Nevertheless, the relic inputs share the final pixel dimensions, not a uniform native angular resolution.

Chapter 4. Methodology

This chapter describes the classification and similarity-scoring pipeline adapted from Baron Perez et al. (2025) for the extended-source sample introduced in Chapter 3. We first learn image representations through SimCLR pre-training with random structural views, then use PCA, HDBSCAN, visual inspection and SI-SSIM filtering to construct eleven morphology classes (Secs. 4.1–4.2). These labels support supervised fine-tuning of a 50-member ensemble (Sec. 4.3), whose probabilities and uncertainty measures define the similarity-to-radio-galaxy score S_{RG} (Sec. 4.4). We then establish the external-label mappings and evaluation metrics used to compare classifications (Sec. 4.5). Finally, we describe the masking and preprocessing of selected radio-relic cutouts to examine the ensemble's response to diffuse emission (Sec. 4.6).

4.1 Model Pre-training

The SimCLR framework with RSV introduced in Sec. 2.4 was implemented using the GreyResNet18 architecture. GreyResNet18 is a ResNet-18 (He et al., 2016) adapted for single-channel input, with its classification layer removed and an input batch-normalisation (BN) layer whose affine parameters are disabled. Each 128×128 cutout is mapped to a 512-dimensional representation. The projection head maps this representation through a 512-dimensional linear layer, BN, and a rectified linear unit (ReLU), followed by a linear map to 256 dimensions. The detailed architecture is shown in Appendix A.

For each source, the augmentation pipeline independently generated two views using a random rotation through a continuous angle, a random vertical flip, brightness jitter with a factor between 0 and 1.1, and a random resized crop retaining between $90\,\%$ and $100\,\%$ of the image area. RSV subsequently modified the pool of second views before the positive pairs were passed to the encoder.

Random structural view via SI-SSIM. The semi-invariant structural similarity (SI-SSIM) between two images is the maximum structural similarity index (SSIM; Wang et al., 2004) obtained over the eight square-image symmetry transforms of the second image: rotations by multiples of 90°, with and without a reflection. This construction is invariant to those eight transforms, rather than to arbitrary orientations, while preserving sensitivity to image structure. SI-SSIM is not scale invariant. A symmetric SI-SSIM matrix is precomputed once from cutouts downsampled to 64×64 pixels and is reused for both RSV and the label-construction procedure of Sec. 4.2.

For a source i, the stored matrix is first restricted to the current minibatch and sorted by decreasing SI-SSIM. The implemented candidate set is

$$\mathcal{C}_i = \{i, n_{i,1}, n_{i,2}, n_{i,3}\}, \qquad |\mathcal{C}_i| = k_{\mathrm{RSV}} = 4.$$

The first candidate is the original source because the matrix diagonal is one. The remaining three candidates are different sources in the minibatch rather than augmentations of the query. One candidate is drawn uniformly, and the query's second augmented view is replaced by the independently augmented second view associated with that candidate. Selecting the self-candidate therefore retains a standard same-source SimCLR pair, whereas selecting one of the other three candidates forms a cross-source positive pair. The nominal probability of forming a cross-source pair is $(k_{\mathrm{RSV}} - 1)/k_{\mathrm{RSV}} = 3/4$. Figure 6 illustrates this selection rule and the shared SimCLR path.

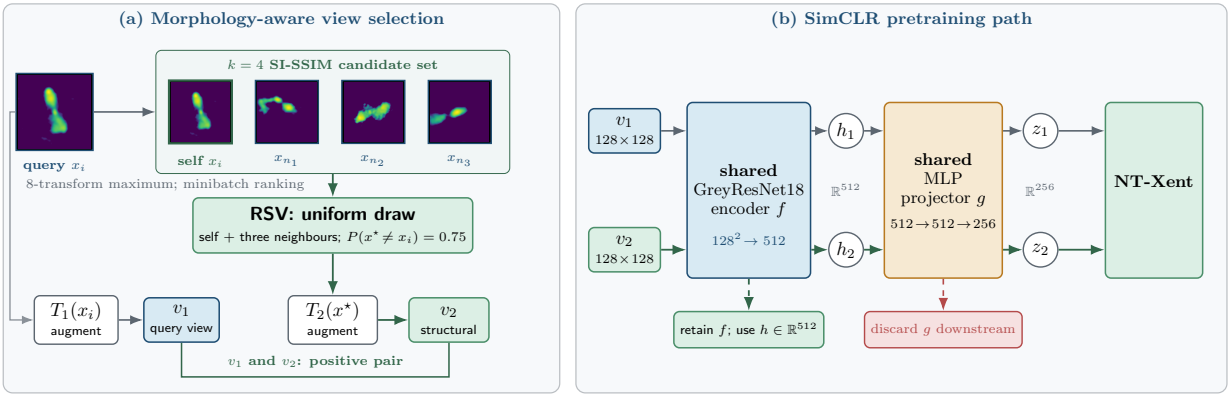


Figure 6: SimCLR pre-training with random structural views (RSV), adapted conceptually from Baron Perez et al. (2025). (a) RSV uses an independently augmented view of either the query or one of its three nearest SI-SSIM neighbours in the minibatch as the second view. (b) Both views pass through the shared encoder f and projector g. NT-Xent is evaluated in the 256-dimensional projection space; the 512-dimensional encoder representation is retained downstream.

The encoder and projector were trained for 205 **epochs** with a **batch size of** 512, AdamW (Loshchilov & Hutter, 2019), a base **learning rate of** 3×10^{-4}, and a **weight decay of** 10^{-4}. A five-epoch linear warm-up was followed by cosine annealing to one fiftieth of the base rate; training used mixed-precision arithmetic and gradient clipping at unit norm. The projection head was discarded after pre-training, and all downstream steps used the 512-dimensional encoder representations.

Baron Perez et al. (2025) report that removing RSV prevented the recovery of one-third of the reported morphology classes. We therefore applied RSV directly without independently assessing its effect.

The encoder and projection head were optimised with the NT-Xent objective defined in Eq. 2.1. Its temperature, $\tau = 0.15$, was selected empirically from a sweep of twenty runs spanning 0.05–1.05. Model selection jointly favoured runs that required more principal components to retain $90\,\%$ of the representation variance, produced more HDBSCAN clusters, and assigned fewer sources to noise. The selected run retained 15 **components**, yielded 59 **non-noise clusters**, and assigned $27\,275$ sources to the noise label.

For the clustering-based evaluation described above, the ℓ_2-normalised representations of all $32\,514$ pre-training sources were reduced by principal component analysis (Jolliffe & Cadima, 2016). HDBSCAN (Campello et al., 2013; McInnes et al., 2017) was then applied with a minimum cluster size of 19, a minimum-samples value of 1, and the Manhattan metric. Of the full sample, $5\,239$ sources ($16.1\,\%$) received non-noise labels. The non-noise cluster sizes range from 19 to $1\,502$ sources and are strongly imbalanced (Figure 7). The three largest clusters, labels 52, 55, and 58, contain $3\,273$ sources, or $62.5\,\%$ of all non-noise memberships.

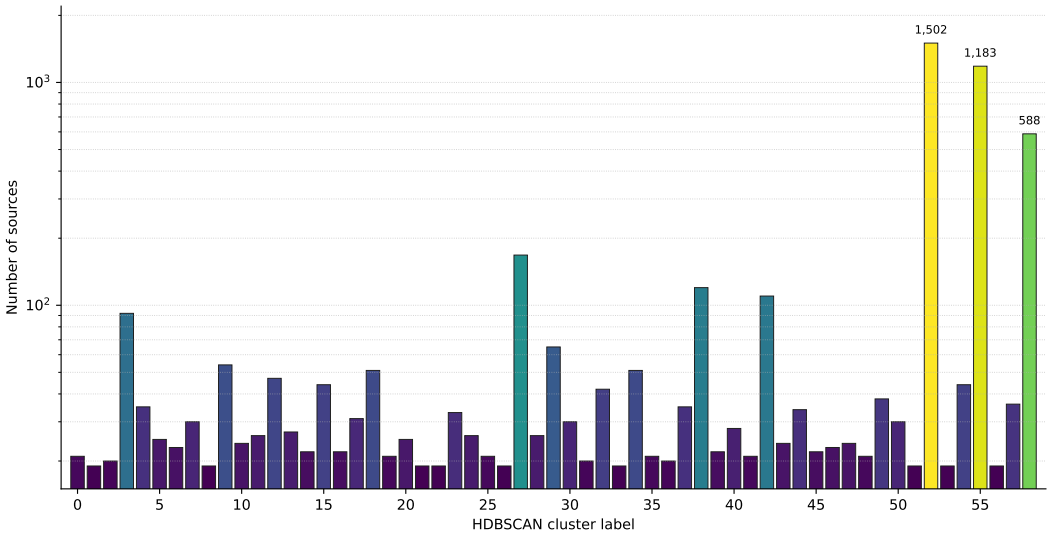


Figure 7: Membership of the 59 non-noise HDBSCAN clusters obtained using the 15 principal components that retain 90 % of the variance. The vertical axis is logarithmic, and the colour scale encodes cluster size.

Figure 8 shows four randomly drawn members from six raw clusters, selected to illustrate the range of compact, elongated, bent, and multi-component structures. These (pre-)labels denote density-based groups in the learned representation space rather than physical morphology classes. The complete raw-cluster overview, including the noise group (cluster label -1), is provided in Appendix B.

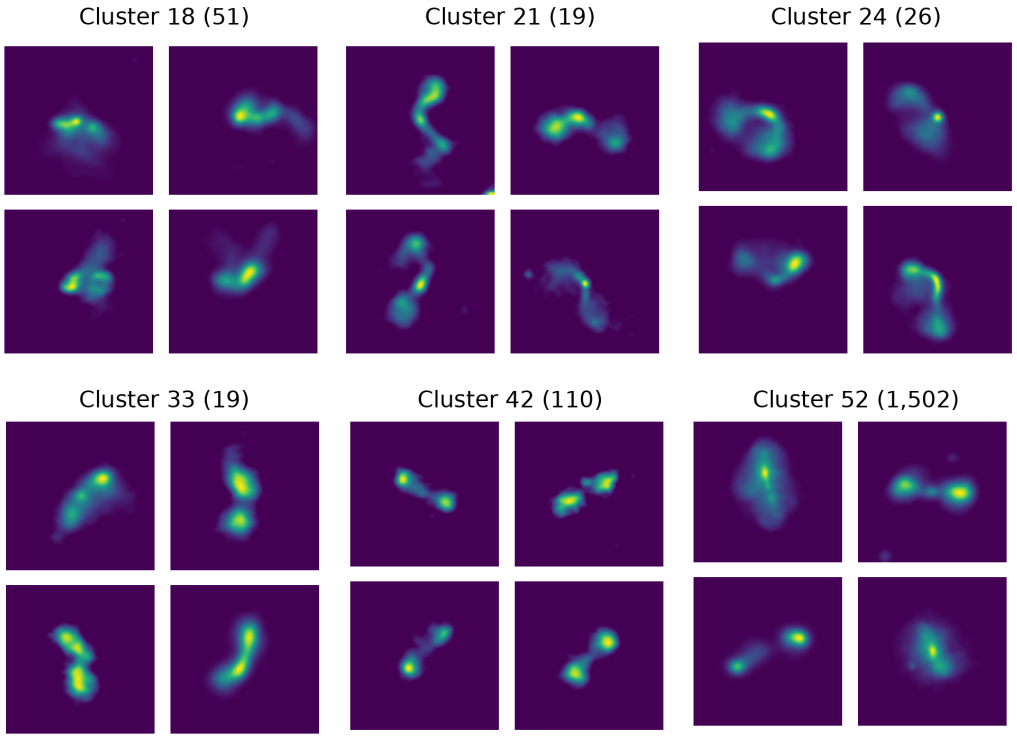


Figure 8: Six raw HDBSCAN clusters. The headings are the original HDBSCAN labels with the total membership of each cluster.

4.2 Cluster-based class construction

At this stage, we derive the internal taxonomy from the clustered representations that form the basis of the labels used to fine-tune the classifier.

First, each raw cluster is inspected visually. Clusters representing the same visual morphology are merged, while clusters unsuitable for the internal taxonomy are returned to the unassigned pool. This consolidation assigns $3\,456$ sources to the eleven morphology classes defined below:

- **Amorphous:** diffuse, extended emission without clearly defined jets, lobes, or hotspots.
- **Asymmetric core-jet (ACJ):** a bright central component with predominantly one-sided extended emission.
- **Bright-core:** compact, centrally peaked emission dominated by a bright core, with faint or absent extended structure.
- **Centre-bright (FRI):** two-sided, FRI-like emission concentrated towards the centre and fading outwards.
- **Core-Displaced:** extended emission containing a dominant compact component offset from the centroid of the resolved structure.
- **Edge-bright (FRII) linear:** approximately collinear, FRII-like emission with brightness peaks near the outer edges.
- **Hybrid (FRX) unclear:** mixed or ambiguous brightening patterns that do not admit a clear FRI or FRII assignment.
- **Large FRII:** large-angular-extent, edge-bright FRII-like emission.
- **Spirals:** extended disc- or spiral-like radio emission rather than a jet-and-lobe morphology.
- **Structured-Bent:** resolved multi-component emission following a curved or bent axis.
- **Unresolved:** point-like or marginally resolved emission in the normalised cutout.

These classes are operational descriptions of appearance in the normalised LoTSS-DR2 cutouts and do not constitute independently verified physical classifications.

Next, to reduce the propagation of atypical cluster members into the classifier, we compute the mean SI-SSIM of each source relative to the other members of its assigned class. Figure 10 shows the seven highest-ranked sources in each class before filtering, together with their mean SI-SSIM values. The most mutually similar $80\,\%$ are retained by default, with class-specific retention fractions between 60 and $90\,\%$ selected according to visual homogeneity. The resulting labelled subsample is presented in Table 5. The class-balanced split, shared by all ensemble members, assigns $2\,602$, 22 and 33 sources to training, validation and testing, respectively. The smallest class is divided approximately $70/15/15$. Every larger class contributes the same numbers of validation and test sources, with all remaining members used for training. This produces class-balanced validation and test sets, although their small sizes prevent reliable estimation of per-class performance. The validation and test sources also participated in self-supervised pre-training. Sources outside the labelled subsample enter the classifier only during inference (Sec. 5.1.1).

Class	Sources	Retained (%)
Amorphous	17	80
Asymmetric core-jet (ACJ)	38	80
Bright-core	946	80
Centre-bright (FRI)	267	80
Core-Displaced	16	80
Edge-bright (FRII) linear	437	80
Hybrid (FRX) unclear	67	90
Large FRII	72	80
Spirals	96	80
Structured-Bent	268	90
Unresolved	433	60
Total	2 657	

Table 5: Labelled subsample produced by cluster-based label construction: sources retained per class and class-specific SI-SSIM retention fraction.

Figure 9 visualises how the curated morphology classes are arranged within the pre-trained representation space and relative to the unassigned sources. Their partial concentration shows that the encoder captures morphology-related structure, while overlaps indicate that the first three PCs, which retain $48.81\,\%$ of the curated-subset variance, do not fully separate the classes. This descriptive projection was not used for class construction.

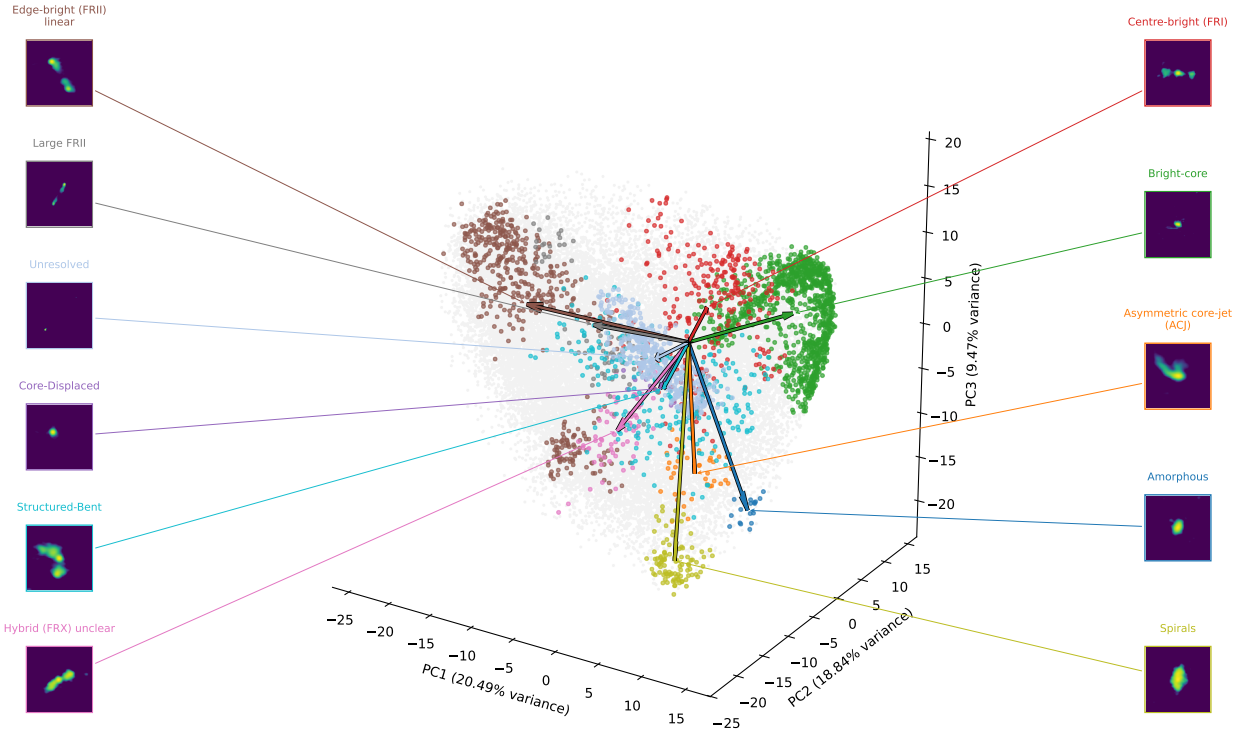


Figure 9: Three-dimensional PCA projection of the curated morphology classes and unassigned sources.

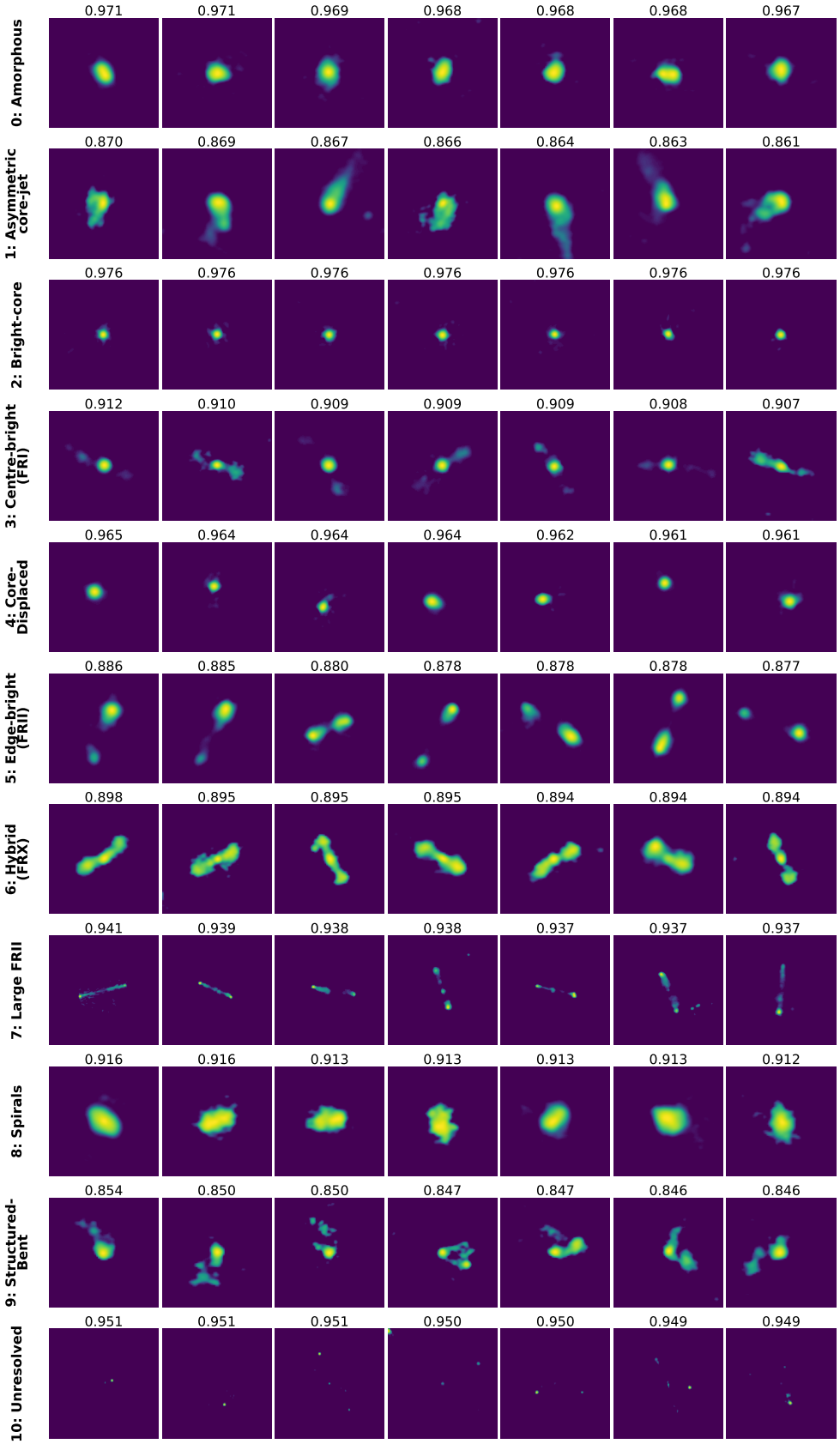


Figure 10: Representative LoTSS-DR2 cutouts for the eleven morphology classes. Each row shows the seven pre-filter sources with the highest mean within-class SI-SSIM.

4.3 Supervised classifier and deep ensemble

At this stage, with the curated morphology classes already defined, the projector is discarded and a classification head is attached to the pre-trained encoder. The classifier maps the 512-dimensional representation to eleven class logits[2]. We fine-tune both the encoder and the head on the labelled subsample.

The fine-tuning loss function (for a minibatch of B sources) combines two terms:

- **Cross-entropy** loss $\mathcal{L}_{\mathrm{CE}}$, which rewards correct class predictions (Salakhutdinov & Hinton, 2007):

$$\mathcal{L}_{\mathrm{CE}} = -\frac{\sum_{i=1}^{B} w_{y_i} \log p_{i,y_i}}{\sum_{i=1}^{B} w_{y_i}}, \tag{4.1}$$

  where p_{i,y_i} is the predicted probability of source i's assigned class y_i. The inverse-frequency weights are $w_c = N_{\mathrm{train}}/(K n_c)$, with N_{train} training sources, n_c members of class c, and $K = 11$ classes.

- **Soft Nearest Neighbour** loss $\mathcal{L}_{\mathrm{SNN}}$, which encourages sources of the same class to have similar representations (Frosst et al., 2019):

$$\mathcal{L}_{\mathrm{SNN}} = -\frac{1}{B}\sum_{i=1}^{B} w_{y_i} \log \frac{\sum_{j\neq i,\, y_j = y_i} \exp(s_{ij}/\tau_{\mathrm{SNN}})}{\sum_{j\neq i} \exp(s_{ij}/\tau_{\mathrm{SNN}})}, \tag{4.2}$$

  where s_{ij} is the cosine similarity between encoder representations, as defined in Sec. 2.4. The neighbour sums run over the current minibatch, and the SNN expression assumes at least one same-class neighbour for each source.

Thus, the total loss is

$$\mathcal{L} \;=\; \alpha\,\mathcal{L}_{\mathrm{CE}} \;+\; (1-\alpha)\,\mathcal{L}_{\mathrm{SNN}}, \tag{4.3}$$

with $\alpha = 0.5$ and SNN temperature $\tau_{\mathrm{SNN}} = 0.07$. We train for 100 epochs with Adam, a batch size of 512, and an initial learning rate of 10^{-3}, reduced when validation accuracy plateaus. Training and validation use random rotations within $\pm 180^{\circ}$ and random vertical flips, whereas testing uses unaugmented images. We retain the model with the highest validation accuracy.

Fine-tuning is repeated $M = 50$ times from the same pre-trained encoder checkpoint to construct a deep ensemble (Lakshminarayanan et al., 2017). At inference, we average the members' class-probability vectors for each of the $32\,514$ catalogue sources. The mean probabilities, agreement A, predictive entropy H, and mutual information MI defined in Eqs. 2.2–2.4 enter the similarity score detailed in Sec. 4.4. Figure 11 summarises this inference path.

[2]Unnormalised class scores converted to probabilities by the softmax function.

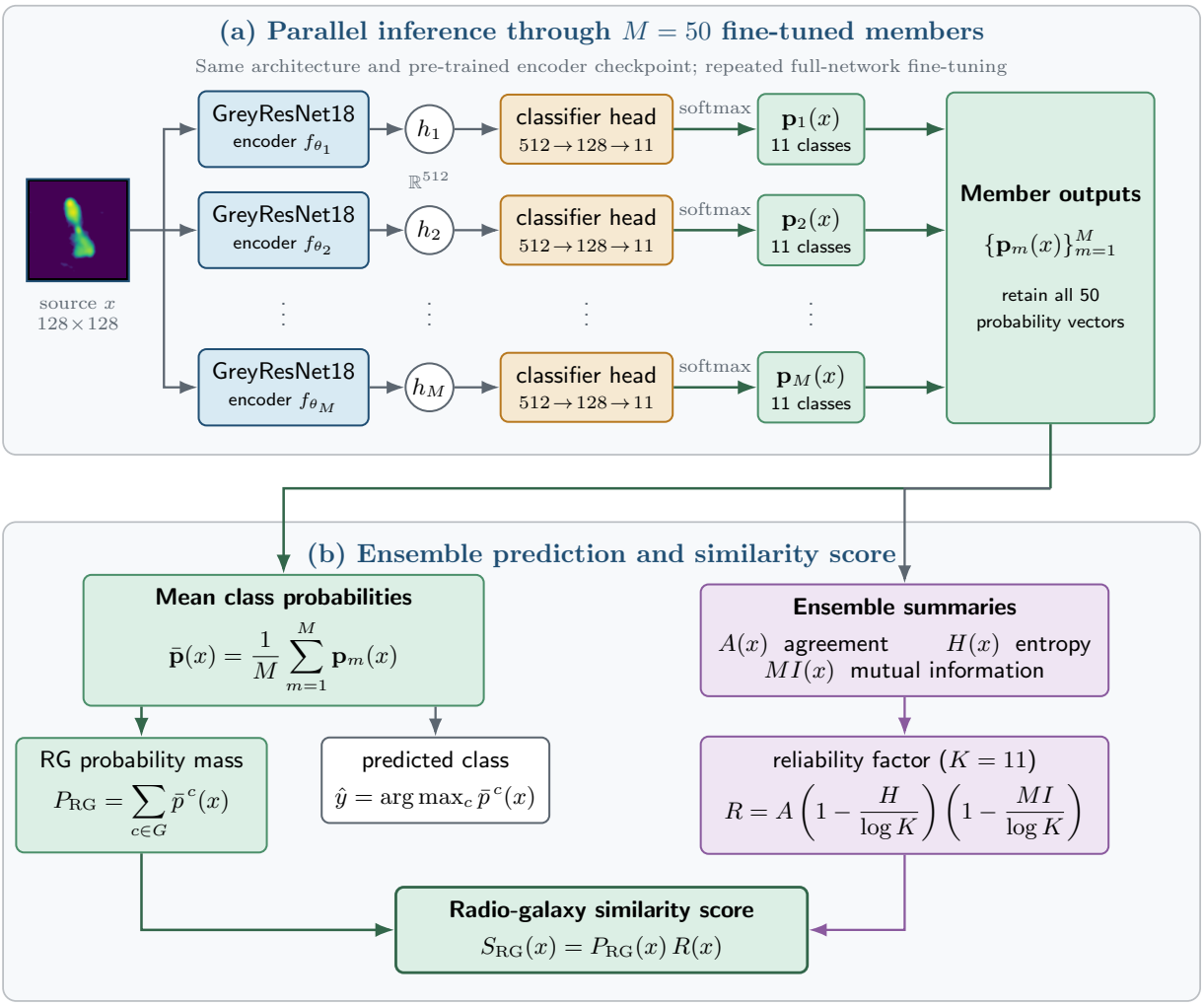


Figure 11: Deep-ensemble inference and similarity scoring.

4.4 Similarity-to-radio-galaxy score S_{RG}

Since the primary objective is not to reproduce a fixed external taxonomy but to quantify how strongly a source resembles the radio-galaxy morphologies represented by the model, we group the internal classes into *radio-galaxy-like* (G) and *non-target* (N) sets.

Using the internal model labels, the radio-galaxy-like set is defined as

$$
\begin{aligned}
G = \{&\text{Asymmetric core-jet (ACJ)},\\
&\text{Bright-core},\\
&\text{Centre-bright (FRI)},\\
&\text{Core-Displaced},\\
&\text{Edge-bright (FRII) linear},\\
&\text{Hybrid (FRX) unclear},\\
&\text{Large FRII},\\
&\text{Structured-Bent}\},
\end{aligned}
$$

and the complementary set is

$$N = \{\text{Amorphous, Spirals, Unresolved}\}.$$

We define these groups operationally from the chosen visual classes, including core-dominated appearances in G. Visual inspection of the optical overlays showed that the majority of the sources in the Spirals class are disc galaxies that exhibit radio emission associated with the galactic disc. Amorphous sources lack sufficiently distinct structure for this target, and Unresolved sources provide insufficient spatial information; either class may still contain physical radio galaxies. This grouping leads to the definition of the radio-galaxy probability mass P_{RG}. With $\mathbf{p}_m(x)$ denoting the probability vector predicted for source x by ensemble member m, as introduced in Eq. 2.2, the radio-galaxy probability mass is

$$P_{\mathrm{RG}}(x) = \sum_{c \in G} \bar{p}^{\,c}(x), \qquad P_{\mathrm{RG}}(x) \in [0, 1]. \tag{4.4}$$

This term measures only how much ensemble-mean probability is assigned to radio-galaxy-like classes. It does not capture the concentration of the probability vector or the consistency of the member predictions. Therefore, three complementary summaries – model agreement $A(x)$ (Eq. 2.2), predictive entropy $H(x)$ (Eq. 2.3), and mutual information $MI(x)$ (Eq. 2.4) – are combined into the definition of a reliability factor,

$$R(x) = A(x) \left(1 - \frac{H(x)}{\log K}\right) \left(1 - \frac{MI(x)}{\log K}\right). \tag{4.5}$$

Dividing H and MI by $\log K$ normalises them by their theoretical maxima for K classes. All three factors, and therefore R, lie in the range $[0, 1]$. Hence, the final *similarity-to-radio-galaxy* score is

$$S_{\mathrm{RG}}(x) = P_{\mathrm{RG}}(x)\, R(x), \qquad S_{\mathrm{RG}}(x) \in [0, 1]. \tag{4.6}$$

For a fixed model and preprocessing pipeline, the score is computed on the same $[0, 1]$ scale for every source. Large S_{RG} values indicate strong support for the chosen radio-galaxy-like model classes together with high ensemble consistency.

Fine-class ambiguity also reduces the score when all probability lies within G. If every member assigns 0.51 to FRI and 0.49 to FRII, then $P_{\mathrm{RG}} = A = 1$ and $MI = 0$, but $S_{\mathrm{RG}} \simeq 0.711$. This design favours concentrated support for a particular learned morphology.

The reliability terms are empirically correlated (see Table 6), but they summarise different aspects of the ensemble output. Agreement A uses only top-class votes, H measures how broadly the ensemble-mean probability is distributed across classes, and MI captures disagreement between the members' probability distributions. Agreement tends to decrease as total or epistemic uncertainty increases, while H and MI increase together when ensemble predictions become less stable. Their multiplicative combination is a design choice.

	A	H	MI
A	1.00	-0.62	-0.52
H	-0.62	1.00	0.49
MI	-0.52	0.49	1.00

Table 6: Empirical Pearson correlations between the reliability terms used in $R(x)$, computed from the classified LoTSS-DR2 catalogue.

4.5 External catalogue mapping and evaluation

As we discussed in Sec. 2.2, there is no single universally adopted taxonomy covering all of these morphologies. Therefore, we adopt the morphological correspondences in Table 7 for the comparisons in Sec. 5.2, rather than assuming exact equivalence between taxonomies. In particular, one-sided ridgelines, X-shaped sources, winged sources, and relaxed doubles only approximately correspond to ACJ, Hybrid, Bent, and Amorphous, respectively. We retain Clews et al.'s uncertain FRI and FRII labels within their respective groups. For external comparison, we combine Edge-bright FRII and Large FRII into a single FRII group.

Clews's one-sided ridgelines are an imperfect ACJ stand-in: inspection of 100 examples suggested that only about $60\,\%$ were genuinely one-sided sources (Clews et al., 2025). Our ambiguous Hybrid class is also broader than a physical FRI/FRII hybrid or an X-shaped morphology. These mappings support comparisons of related appearances, not interchangeable physical labels.

For Horton's multi-label catalogue, we assign the first class satisfying its conditions in the order FRI, FRII, Spirals, Amorphous, Hybrid, ACJ, then Bent. Sources satisfying none of these rules remain unmapped; multiple flags do not automatically imply exclusion.

Class ID	Class name	Clews et al. (2025)	Horton et al. (2025)	Chilufya et al. (2025)
0	Amorphous	–	`relaxed`[a], `diffuse`[a]	relaxed
1	ACJ	one_sided_ridge	`onesided`	–
2	Bright-core	–	–	–
3	FRI	FRI, uncertain_FRI	`fri`[b]	FRI
4	Core-Displaced	–	–	–
5	FRII	FRII, uncertain_FRII	`frii`[b]	FRII
6	Hybrid	–	`hybrid`, `xshaped`	–
7	Large FRII	–	–	–
8	Spirals	–	`spiral`	–
9	Bent	bent	`cshaped`[a], `sshaped`[a], `wings`[a]	WAT, NAT, headtail
10	Unresolved	–	–	–
Unmapped: Clews – *other*, *triple*; Horton – no mapping rule satisfied.				

[a] Only when `fri` and `frii` are both false.

[b] Only when the other FR flag is false (exclusive).

Table 7: Adopted morphological correspondences between reference catalogues and our internal classes. A dash (–) indicates that no direct mapping is adopted.

For each external-catalogue comparison, the target morphology defines the positive class and the remaining mapped morphologies define the negative class. A true positive (TP) or true negative (TN) agrees with the reference assignment. A false positive (FP) is predicted positive but is negative in the reference, while a false negative (FN) is predicted negative but is positive in the reference. These four counts form the confusion matrix. Precision measures the purity of the predicted positive sample, whereas recall measures the completeness of the positive class:

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{4.7}$$

The F_1 score combines precision and recall through their harmonic mean,

$$F_1 = 2\,\frac{\mathrm{precision}\,\mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}. \tag{4.8}$$

A higher F_1 score indicates better classification performance on the evaluated sample.

We estimate uncertainty in the reported metrics from the number of evaluated sources and repeated resampling of each comparison sample. The resulting bounds describe how the metrics may vary with sample membership – label errors, selection bias and training variability require separate assessment. Each catalogue is evaluated separately, with calculation details in Appendix D.

We also repeat the comparisons after excluding all $2\,657$ curated sources, across angular-size ranges, and without Clews's uncertain FR labels. Excluding curated sources removes overlap with supervised training, although the remaining sources still participated in pre-training.

4.6 Application to diffuse radio emission

We used the selected radio relics described in Sec. 3.3 to test the trained ensemble on diffuse radio emission. The aim is to examine how their morphology is represented by the radio-galaxy classifier. Bright radio galaxies within a relic cutout can dominate the normalised input, while clipping can suppress the relic's low-surface-brightness emission. We therefore mask neighbouring sources before preprocessing to focus the test on the diffuse emission.

We start with relic-centred LoTSS-DR3 cutouts with a side length of $1.5 \times \mathrm{LAS}$. Then, we manually draw polygonal regions around the relic emission, exclude pixels outside them, estimate the sigma-clipped standard deviation σ from the retained pixels and then set both masked pixels and values below 3σ to this floor. The images are resampled to 128×128 pixels and min–max normalised for inference.

These selected examples illustrate the model's response to isolated relic emission, rather than its performance on unmasked fields. The assigned labels describe resemblance to the learned classes.

Chapter 5. Results

In this chapter we present the morphology catalogue and similarity scores produced by the trained ensemble. First, the internal class assignments and score distributions are described; then we compare the predictions with external morphology catalogues. Finally, we examine the model response to selected radio relics.

5.1 Internal classification and S_{RG} evaluation

The internal results summarise the ensemble predictions across the full LoTSS-DR2 sample. Class membership, confidence and agreement are presented first, followed by the distributions of the similarity-score components and membership in the adopted score bands.

5.1.1 Ensemble classification catalogue

The ensemble classification catalogue contains $32\,514$ sources across eleven morphology classes (Table 8). Edge-bright FRII and Structured-Bent are the most common assignments, together accounting for $45.4\,\%$ of the sample, whereas Amorphous sources constitute only $1.2\,\%$. These are predicted image-morphology groups, rather than independently confirmed physical source populations.

Bright-core sources have the highest mean confidence, while Amorphous and Asymmetric core-jet sources have the lowest (Table 8). Confidence and agreement describe the concentration and consistency of predictions, rather than classification accuracy. Figure 12 compares the most and least confident assignments within each class.

The ensemble correctly classifies 28 of the 33 test images withheld from supervised training ($84.8\,\%$). Most errors involve ACJ and Structured-Bent sources (Table 17). With only three examples per class and prior exposure during pre-training, this test provides limited evidence of performance on new survey sources.

Class	Count	Fraction	Confidence	Agreement
Amorphous	390	1.2 %	0.651	0.869
Asymmetric core-jet (ACJ)	1 976	6.1 %	0.660	0.894
Bright-core	3 728	11.5 %	0.865	0.946
Centre-bright (FRI)	3 323	10.2 %	0.785	0.912
Core-Displaced	980	3.0 %	0.697	0.834
Edge-bright (FRII) linear	7 574	23.3 %	0.821	0.931
Hybrid (FRX) unclear	2 865	8.8 %	0.718	0.909
Large FRII	2 425	7.5 %	0.774	0.876
Spirals	963	3.0 %	0.756	0.907
Structured-Bent	7 197	22.1 %	0.740	0.918
Unresolved	1 093	3.4 %	0.711	0.873
Total	32 514	100 %		

Table 8: Class composition and mean ensemble confidence and agreement.

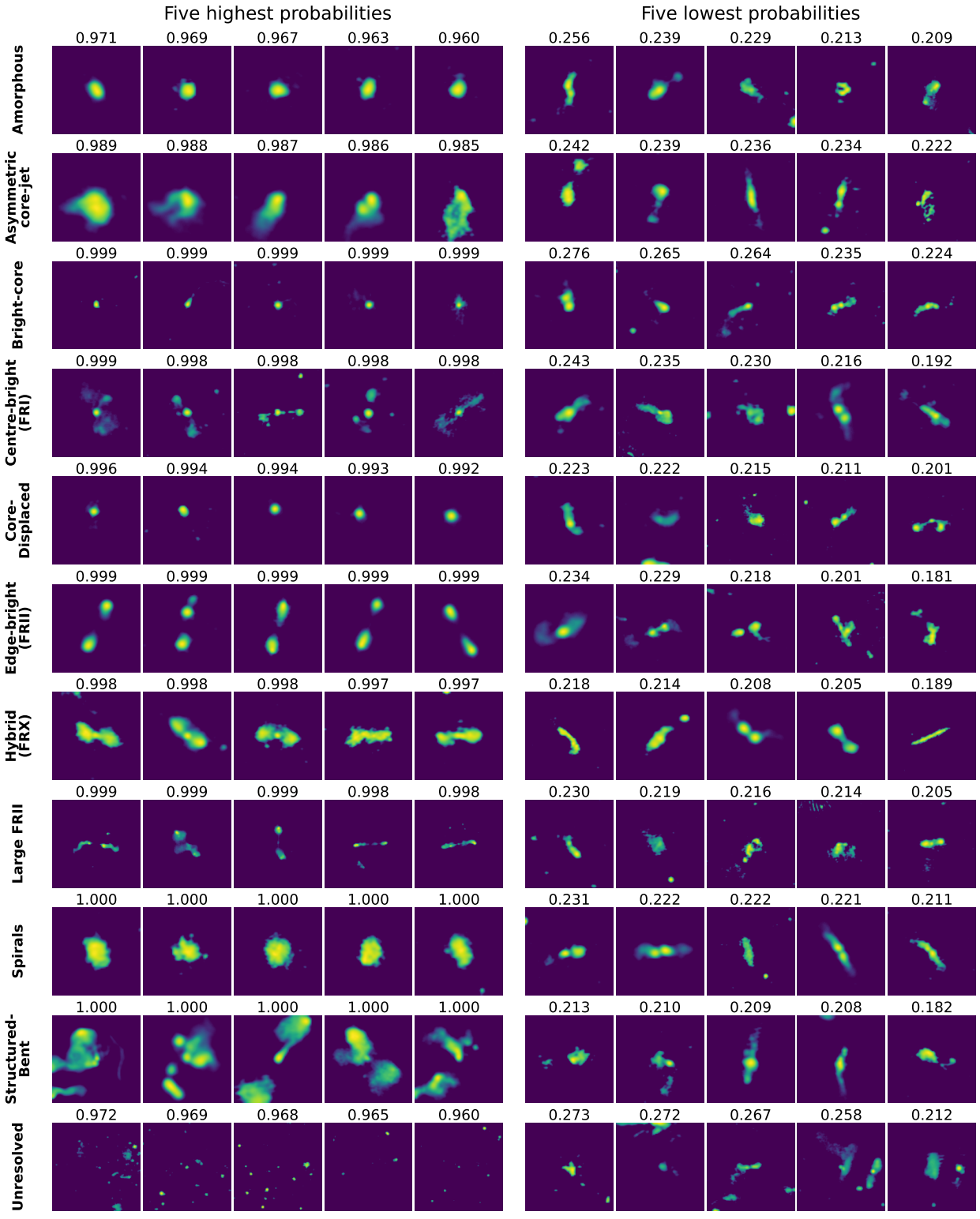


Figure 12: Five highest-confidence (left) and five lowest-confidence (right) sources per predicted class. Values above cutouts are ensemble-mean probabilities of the assigned class.

5.1.2 Score distributions and operational bands

Figure 13 shows how the class probabilities and ensemble consistency combine into S_{RG}. The radio-galaxy probability mass P_{RG} is concentrated near unity, while the reliability factor R is more broadly distributed. A large probability mass in the radio-galaxy-like classes therefore does not necessarily produce a high final score.

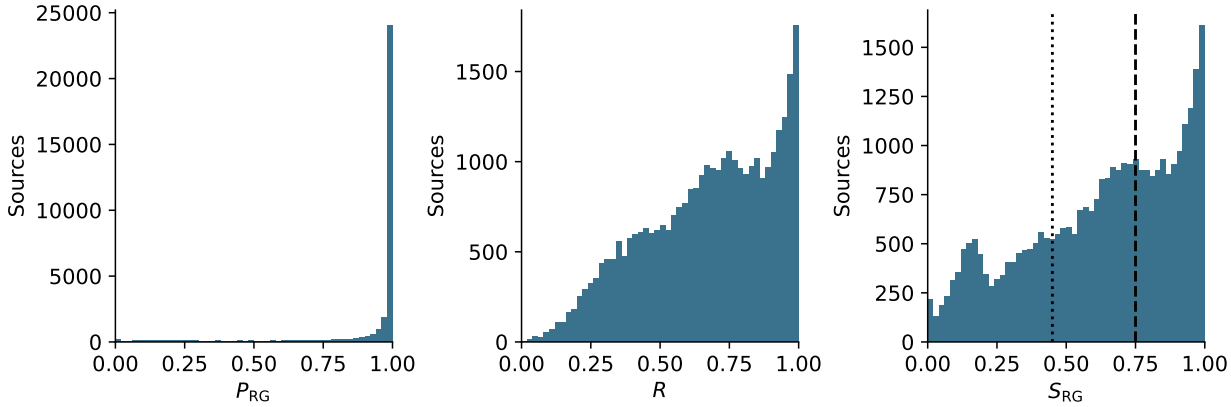


Figure 13: Score components for all $32\,514$ sources. The dashed and dotted lines in the right panel mark $S_{\mathrm{RG}} = 0.75$ and 0.45, respectively.

Table 9 divides the catalogue into the adopted high, transitional and low similarity bands. The high-score band is the largest of these review subsets.

Band	Count	Fraction
High: $S_{\mathrm{RG}} \geq 0.75$	12 875	39.6 %
Transitional: $0.45 \leq S_{\mathrm{RG}} < 0.75$	10 956	33.7 %
Low: $S_{\mathrm{RG}} < 0.45$	8 683	26.7 %

Table 9: Catalogue membership in the three operational score bands.

The class-conditioned distributions in Figure 14 reflect both the adopted G/N grouping and the ensemble's support for individual predictions. The non-target classes favour low scores; within G, Bright-core has the highest median and Core-Displaced the lowest. The substantial spread within classes shows why a class assignment alone does not determine the review band.

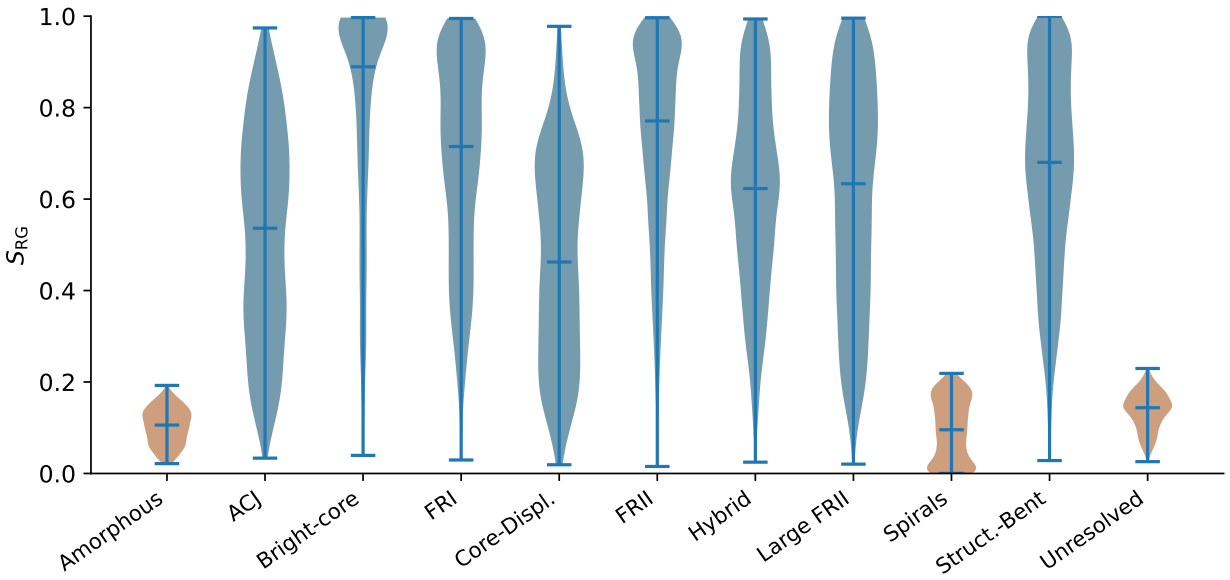


Figure 14: Score distributions by ensemble-assigned class. Blue and orange denote the G and N groups; central horizontal marks show medians.

5.2 Comparison with external morphology catalogues

We compare the ensemble assignments with the mapped labels of Clews et al. (2025), Horton et al. (2025), and Chilufya et al. (2025), using the correspondences in Sec. 4.5. The mapped samples in Table 2 include 645, 718, and 198 members of the internally curated subsample, respectively. These comparisons assess agreement with independently compiled label schemes, not an established physical ground truth. FRII includes both Edge-bright FRII and Large FRII throughout the comparison.

The main findings remain after excluding all curated sources (Table 10). The figures retain the full mapped samples; Appendix D gives additional checks. The exclusion removes overlap with supervised training, although these sources still participated in pre-training. This check uses the curated-excluded selections in the final column of Table 2.

Catalogue	Sources	FRI F_1	FRII F_1
Clews	7757	$0.389^{+0.020}_{-0.021}$	$0.735^{+0.013}_{-0.013}$
Horton	8352	$0.281^{+0.020}_{-0.022}$	$0.743^{+0.010}_{-0.010}$
Chilufya	2590	$0.445^{+0.036}_{-0.036}$	$0.455^{+0.032}_{-0.033}$

Table 10: External correspondence after excluding all curated sources. Signed uncertainties give the distances to the $95\,\%$ interval bounds.

Figure 15 shows the class correspondence for all three catalogues. We then examine FRI and FRII separately.

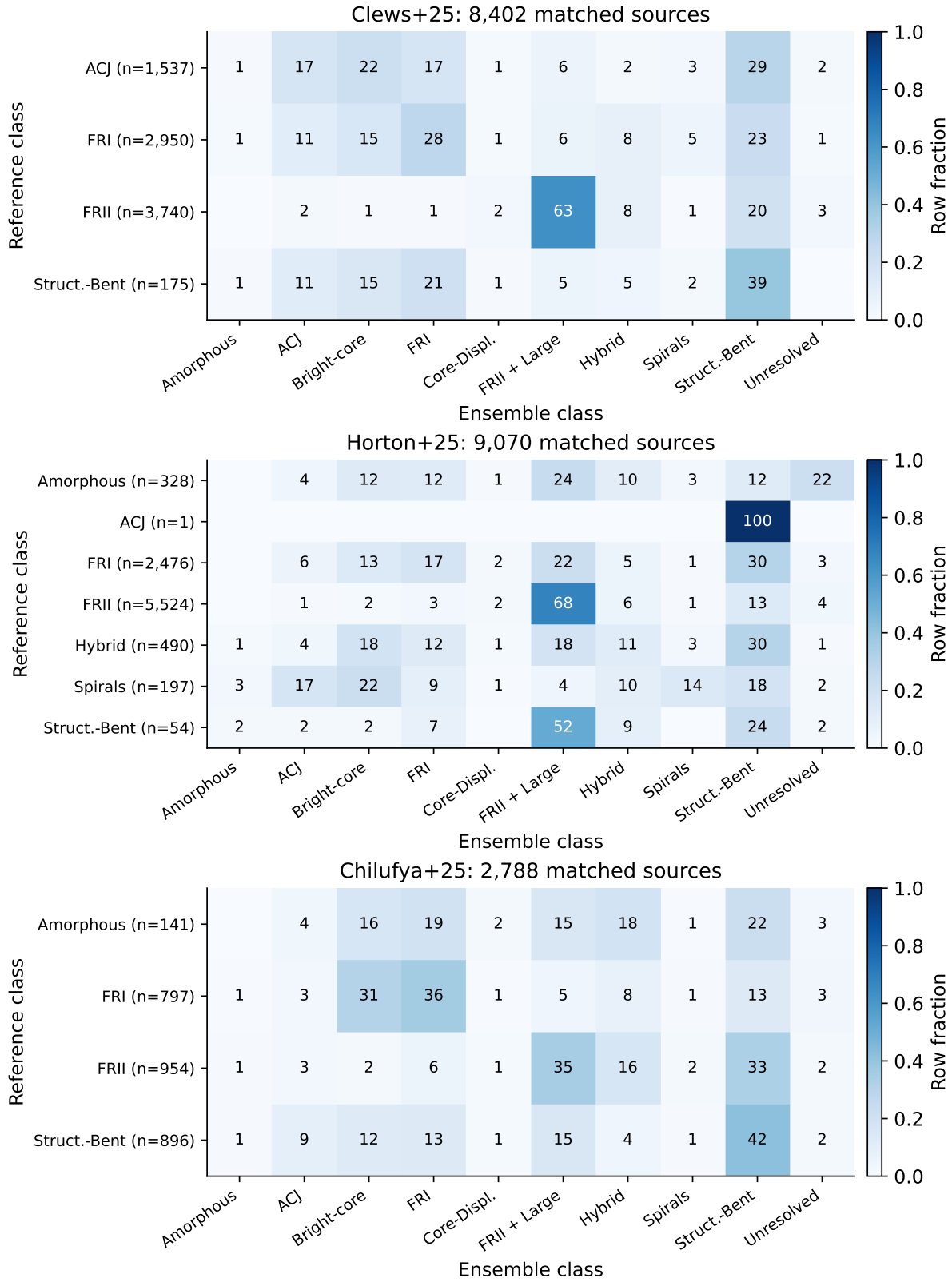


Figure 15: Row-normalised class correspondence for Clews, Horton, and Chilufya. Rows are reference classes with source counts in parentheses; columns are ensemble predictions. Edge-bright FRII and Large FRII predictions are combined. Cells above $0.5\,\%$ are annotated with percentages; reference classes absent from a catalogue are omitted.

5.2.1 FRI correspondence

The model recovers FRI morphology poorly across all three catalogues (Table 11).

Catalogue	Reference sources	TP	FP	FN	F_1
Clews	2950	818	334	2132	$0.399^{+0.019}_{-0.019}$
Horton	2476	431	278	2045	$0.271^{+0.020}_{-0.020}$
Chilufya	797	285	198	512	$0.445^{+0.033}_{-0.035}$

Table 11: FRI agreement with the mapped reference labels; uncertainties indicate $95\,\%$ intervals.

Frequent Bright-core assignments suggest that central brightness dominates over extended jets and lobes, while Structured-Bent assignments reveal sensitivity to bending (Figure 16). These patterns also reflect catalogue-dependent class boundaries, particularly Horton's inclusion of curved sources within FRI (Horton et al., 2025).

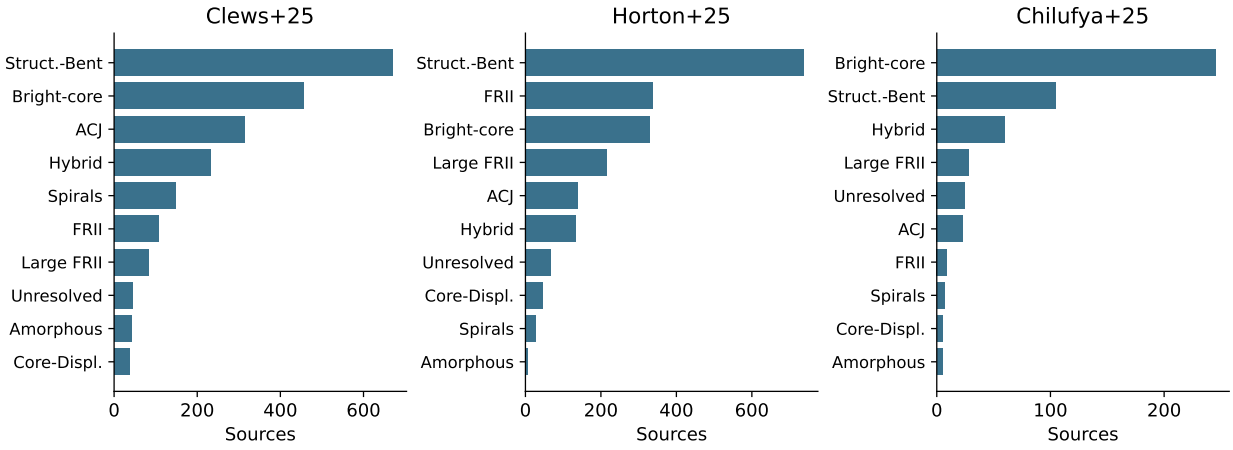


Figure 16: Ensemble assignments of reference FRI sources not predicted as FRI.

The ensemble also assigns alternative morphologies to reference FRI sources with high confidence (Figure 17).

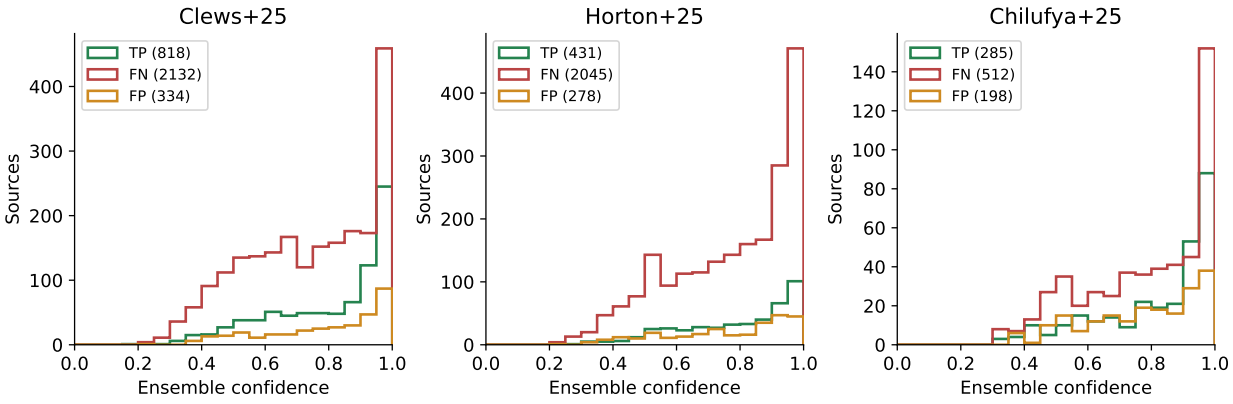


Figure 17: Ensemble confidence for FRI true positives, false negatives, and false positives relative to each reference catalogue.

5.2.2 FRII correspondence

FRII correspondence is strongest against Clews and Horton and weaker against Chilufya (Table 12).

Catalogue	Reference sources	TP	FP	FN	F_1
Clews	3740	2359	287	1381	$0.739^{+0.012}_{-0.012}$
Horton	5524	3744	759	1780	$0.747^{+0.010}_{-0.010}$
Chilufya	954	333	192	621	$0.450^{+0.032}_{-0.032}$

Table 12: FRII agreement with the mapped reference labels; uncertainties indicate $95\,\%$ intervals.

Against Chilufya, FRI and FRII performance is similar, including after curated-source exclusion.

The remaining disagreements concentrate in Structured-Bent and Hybrid across all three catalogues (Figure 18). This pattern suggests that curvature and mixed brightening patterns complicate the model's recognition of edge-brightened FRII structure.

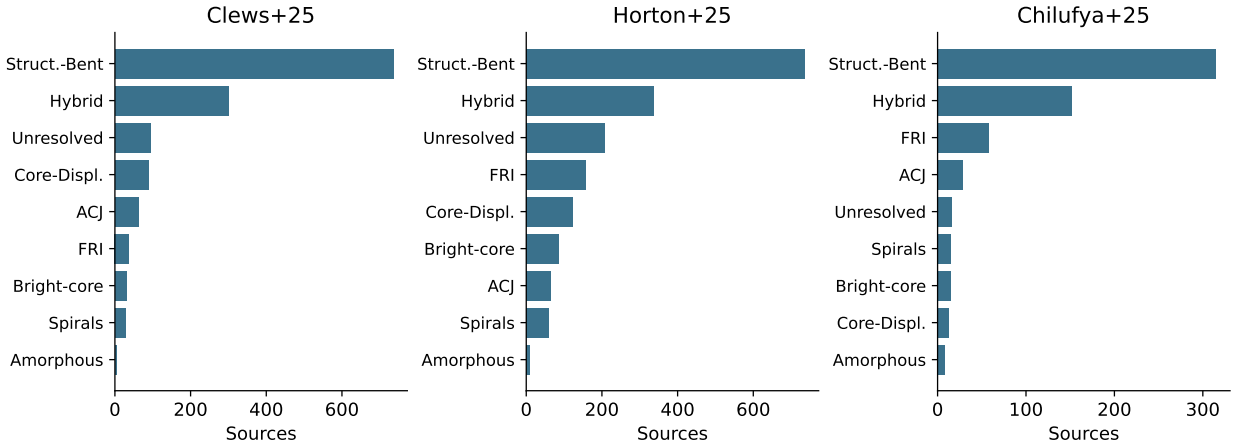


Figure 18: Ensemble assignments of reference FRII sources predicted as neither Edge-bright FRII nor Large FRII.

Reference FRII sources assigned to either internal FRII class have higher median confidence than those assigned to other classes (Figure 19).

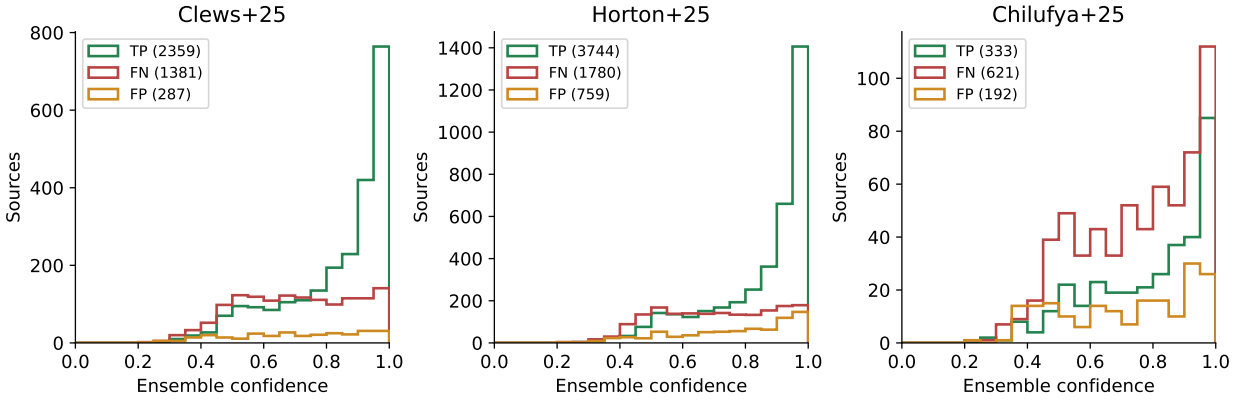


Figure 19: Ensemble confidence for FRII true positives, false negatives, and false positives.

5.2.3 Three-catalogue consensus

Of the sources in the common-source sample defined in Table 3, 311 have usable mapped labels in all three references. Of these, 149 receive the same mapped label, forming the consensus subset independently of the ensemble predictions; ten belong to the curated subsample. The ensemble agrees with the common reference label for fewer than half of these sources (Table 13).

Reference class	Sources	TP	FP	FN	F_1
FRI	62	25	0	37	$0.575^{+0.117}_{-0.136}$
FRII	87	43	4	44	$0.642^{+0.090}_{-0.104}$

Table 13: Agreement with labels shared by all three catalogues; uncertainties indicate $95\,\%$ intervals.

Agreement remains below half after excluding the curated sources. Reference FRI sources receive Bright-core assignments almost as often as FRI assignments, while reference FRII disagreements mainly involve Structured-Bent and Hybrid (Figure 20).

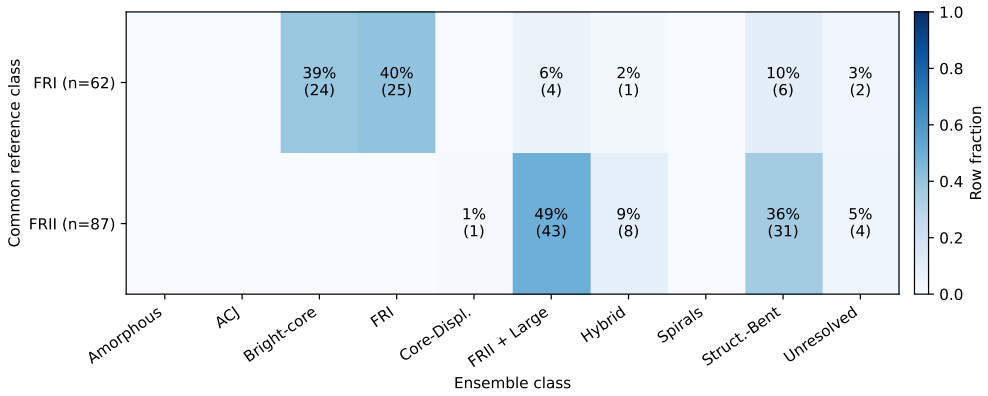


Figure 20: Ensemble assignments for the 149 three-catalogue consensus sources. Cells give row percentages and source counts.

5.2.4 S_{RG} distribution across mapped-reference sources

Table 14 includes all sources with a usable reference label in either G or N. For each catalogue, we separate fine-class agreements from disagreements and report the ensemble's G/N assignments within each subset.

Figure 21 shows the score components for the $16\,177$ study sources with at least one mapped reference label (Table 3). Each source is counted once, irrespective of catalogue overlap or agreement with the ensemble. For sources assigned to N, the P_{RG} distribution is concentrated towards low values, indicating weak support for radio-galaxy-like morphologies. Their scores remain below $S_{\mathrm{RG}} = 0.25$, even where ensemble reliability is high.

Catalogue	Subset	Sources	Share (%)	Assigned to G n (row %)	Assigned to N n (row %)
Clews	All mapped	8 402	100.0	7 943 (94.5)	459 (5.5)
	Agreements	3 501	41.7	3 501 (100.0)	0 (0.0)
	Disagreements	4 901	58.3	4 442 (90.6)	459 (9.4)
Horton	All mapped	9 070	100.0	8 551 (94.3)	519 (5.7)
	Agreements	4 271	47.1	4 244 (99.4)	27 (0.6)
	Disagreements	4 799	52.9	4 307 (89.7)	492 (10.3)
Chilufya	All mapped	2 788	100.0	2 670 (95.8)	118 (4.2)
	Agreements	991	35.5	991 (100.0)	0 (0.0)
	Disagreements	1 797	64.5	1 679 (93.4)	118 (6.6)

Table 14: Fine-class correspondence for all mapped reference sources.

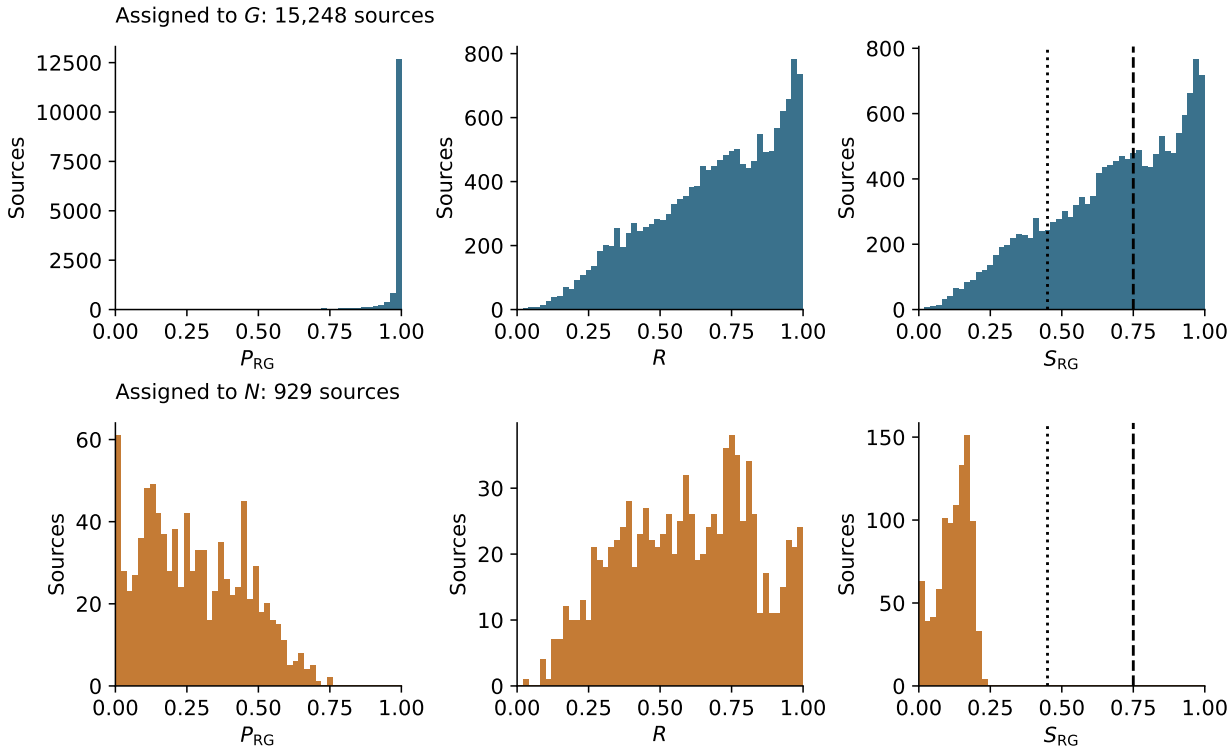


Figure 21: Score components for sources with mapped reference labels, separated by ensemble-assigned G (top) and N (bottom). Dashed and dotted lines mark $S_{\mathrm{RG}} = 0.75$ and 0.45.

For the visual examples in Figure 22, we select sources with at least one mapped reference label and agreement between every available mapped label and the ensemble assignment. We show five examples for each of six internal classes (ACJ, FRI, FRII, Hybrid, Spirals, and Structured-Bent). We prioritise sources covered by more reference catalogues. These examples illustrate agreement without assuming its prevalence. In Appendix C we present the complementary inspection sample (disagreement examples for each of the eleven internal classes).

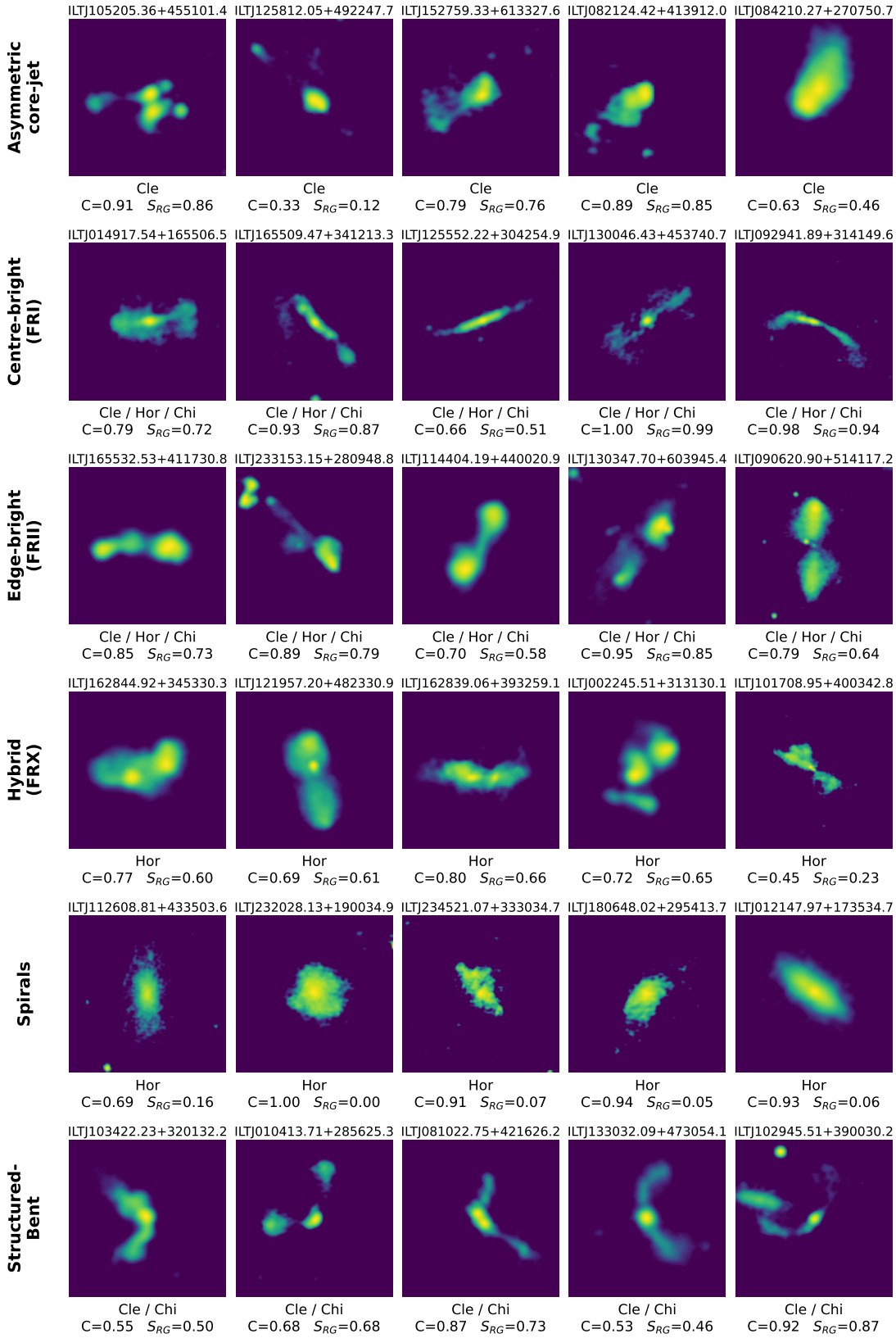


Figure 22: Agreement examples grouped by ensemble class. Cle, Hor, and Chi identify supporting catalogues. C denotes confidence and S_{RG} the morphology score.

5.3 Application to radio relics

In order to test the model's response to the isolated diffuse emission, we apply the 50-member ensemble to the ten selected, polygon-masked relic cutouts listed in Table 4. Detailed preprocessing is described in Sec. 4.6. Figure 23 shows the actual inputs. Identification of relics in unmasked survey fields using an automated approach is a challenging task and lies beyond the scope of this work.

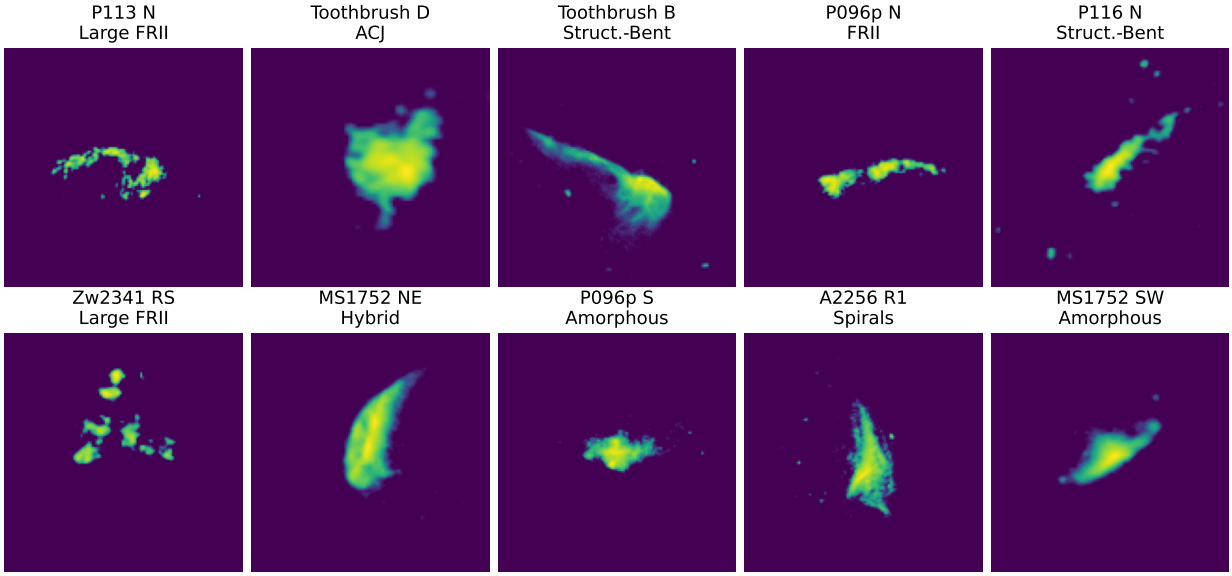


Figure 23: The ten masked relic inputs, shown in decreasing S_{RG} order with their predicted morphology labels.

Relic	Predicted class	C	A	H	MI	P_{RG}	S_{RG}
P113 N	Large FRII	0.892	0.960	0.431	0.108	0.994	0.748
Toothbrush D	ACJ	0.739	1.000	0.863	0.058	0.941	0.588
Toothbrush B	Structured-Bent	0.637	0.980	0.974	0.035	0.991	0.568
MS1752 SW	Amorphous	0.489	0.780	1.436	0.091	0.267	0.080
MS1752 NE	Hybrid	0.590	0.780	0.800	0.074	0.613	0.309
Zw2341 RS	Large FRII	0.676	0.700	0.956	0.443	0.990	0.340
P096p N	FRII	0.689	0.840	0.942	0.195	0.989	0.464
P096p S	Amorphous	0.801	0.960	0.833	0.122	0.180	0.107
P116 N	Structured-Bent	0.523	0.740	1.207	0.049	0.980	0.353
A2256 R1	Spirals	0.887	1.000	0.497	0.014	0.103	0.082

Table 15: Ensemble outputs for the selected relics. C denotes confidence; H and MI are in nats[3].

The relics span seven predicted classes (Table 15). None is assigned to Unresolved. These labels describe resemblance to structures learned from the radio-galaxy sample. Four relics fall in the transitional band and six in the low band; none reaches the high-score band (Figure 24). P113 N has the highest score, just below the high-band boundary.

[3]One nat is the amount of information from an event with a probability of $1/e$.

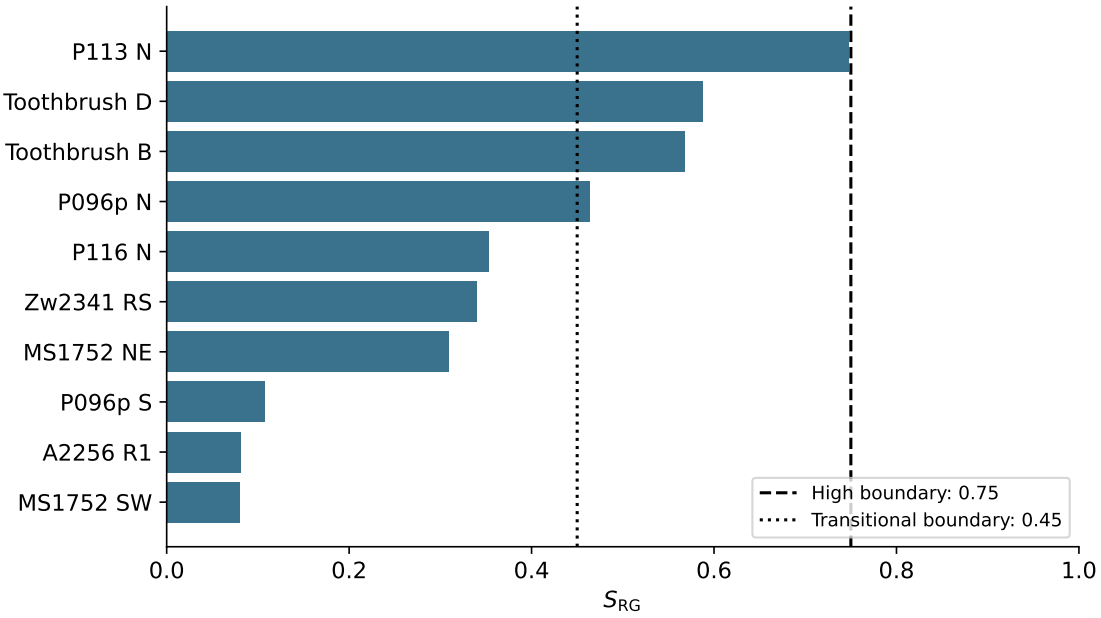


Figure 24: Similarity scores for the ten relics. Dashed and dotted lines mark the high and transitional boundaries, respectively.

For Zw2341 RS, almost all probability lies in G, yet uncertainty and disagreement among members reduce the final score. Conversely, A2256 R1 receives a confident and consistent Spirals assignment, which places it in N and gives a low score. These examples show how low scores can arise either from weaker reliability within the target groups or from concentrated support for a non-target class (Table 15).

The retained arcs and elongated ridges can receive radio-galaxy-like assignments despite their different physical origin. The range of outputs demonstrates why neither high P_{RG} nor low S_{RG} alone identifies a relic. This visually selected sample does not establish relic-detection completeness or out-of-distribution performance, and the results remain conditional on the manual isolation and clipping of diffuse emission.

Chapter 6. Discussion

In this chapter we discuss the model's morphological strengths and weaknesses, the use of its similarity score, and the main limits of the analysis.

6.1 Morphological classification and external correspondence

By extending the approach of Baron Perez et al. (2025), we obtained a catalogue of $32\,514$ sources with $60'' \leq \mathrm{LAS} \leq 1000''$. Its eleven classes provide a common morphological description for selecting and inspecting this extended-source sample.

Baron Perez et al. (2025) classified $42\,230$ sources with LAS between $30''$ and $60''$ into twelve classes. We retain their host-centred image preparation and learning framework, while adopting a larger-angular-size sample and an eleven-class curation. The changed population and taxonomy prevent a direct comparison of classification performance.

We find stronger FRII than FRI correspondence against Clews and Horton, while performance is similar for the two classes against Chilufya (Tables 11 and 12). These comparisons treat Edge-bright FRII and Large FRII as a single morphological family.

Frequent Bright-core assignments among reference FRI sources suggest that central brightness outweighs faint jets and lobes in the learned classification. Structured-Bent assignments suggest sensitivity to curvature, which some reference schemes include within FRI. Reference FRII sources assigned to Structured-Bent or Hybrid likewise suggest difficulty separating edge brightening from curvature and mixed structure (Figures 16 and 18).

The reference catalogues differ in source selection, classification method and available imaging (Clews et al., 2025; Horton et al., 2025; Chilufya et al., 2025). Similar FRI and FRII disagreements persist among the 149 sources whose mapped labels agree across all three catalogues, identifying cases that merit closer inspection (Sec. 5.2.3). Most sources with fine-class disagreements receive a radio-galaxy-like label, although these labels also dominate the reference samples (Table 14).

Image preparation offers possible explanations for some disagreements:

- resampling cutouts to 128×128 pixels can blur small features;
- clipping can suppress faint emission, while linear intensity scaling leaves faint features weak relative to bright components;
- neighbouring sources and host centring can also affect the structure presented to the model.

Thus, Bright-core and Core-Displaced assignments require inspection of the original radio field and host information for physical interpretation.

Unlike Baron Perez et al. (2025), we omitted active-pixel selection because it would substantially reduce the sample. It could also exclude diffuse or extended sources, including some Large FRII sources, if too few pixels exceed the intensity threshold.

Visual cluster consolidation and SI-SSIM filtering help define the class boundaries (Sec. 4.2). Their outcome depends on how carefully clusters are grouped into morphologically coherent classes and which examples are retained.

6.2 Similarity scoring and diffuse-emission applications

The similarity score introduced here adds a filtering layer to the morphology catalogue by combining support for radio-galaxy-like (G) classes with ensemble reliability. Its proposed use is to prioritise sources for inspection within the fitted model and adopted preprocessing.

The model already assigns $92.5\,\%$ of the full sample to G, so frequent G assignments among external disagreements are mainly a descriptive result.

In the mapped-reference sample, sources assigned to the non-target group N favour low P_{RG} and remain below $S_{\mathrm{RG}} = 0.25$, including cases with high reliability (Figure 21). Within G, reliability separates concentrated, consistent predictions from weaker support. Examining P_{RG} and R separately shows which factor reduces the score.

We should note that the score bands depend on the fitted model and the adopted G/N definitions. Independent review is needed to establish their selection quality and any improvement over P_{RG} alone.

The ten selected relics receive seven morphology labels, illustrating how diffuse arcs and ridges can resemble structures learned from radio galaxies (Table 15). All fall in the transitional or low-score bands, despite several having large P_{RG}. These results show that morphological resemblance can extend across physically different source populations.

Manual masking is important for isolating relic emission, while clipping and rescaling determine which features remain visible. The conclusions therefore apply to these deliberately prepared examples. Automated identification of radio relics or other diffuse emission in unmasked survey fields lies beyond this work's scope.

The polygon also changes the pixels used to estimate the clipping threshold, coupling source isolation to intensity preprocessing. P113 N lies just below the high-score boundary; its band assignment should therefore be tested against reasonable changes in masking and clipping.

6.3 Limitations and implications for future work

One limitation of the pipeline is its dependence on the source associations, host coordinates and angular sizes supplied by the value-added catalogue. Errors in these quantities can affect sample membership, cutout centring and the extent of emission included in the model input. Applying the pipeline to another survey requires equivalent source associations, host positions and angular-size estimates.

Another bottleneck of the framework is manual curation, which is time-consuming and depends on the analyst's judgement. Reproducibility requires preserving the original cluster memberships, cluster-to-class mapping and filtering choices; in Appendix B we record the grouping used in this work. Independent repeat curation would test the stability of the class boundaries.

The validation and test sets used for fine-tuning contain only 22 and 33 sources, making performance estimates for individual classes uncertain. Their selection from the curated sample also favours the morphologies retained by the filtering procedure. External comparisons use independently assigned reference labels, although some compared sources were used for classifier training (Sec. 5.2).

The main evaluation priority is an independently reviewed sample covering all classes, score bands and angular sizes. Testing on additional sky fields would assess transfer beyond the population used for pre-training.

Agreement also varies with angular size: against Horton, FRI recovery is weakest among the smallest sources (Table 18). Under the adopted resampling, a $6''$ beam spans approximately $512/\mathrm{LAS}$ pixels, with

LAS expressed in arcseconds, decreasing from 8.53 to 0.512 pixels across the sample's size range and limiting the preservation of small-scale structure in the largest sources. Since morphology and resolution vary together, controlled tests of intensity scaling, source isolation and angular-size dependence would help distinguish their contributions to the observed disagreements, as outlined in Chapter 7.

Chapter 7. Conclusion and Outlook

7.1 Main conclusions

We produced an eleven-class catalogue of $32\,514$ extended LoTSS-DR2 sources using the adapted framework of Baron Perez et al. (2025). Self-supervised pre-training, cluster curation and a 50-member ensemble provide morphology assignments and uncertainty summaries.

FRII correspondence is stronger than FRI correspondence against Clews and Horton, while the two are similar against Chilufya. FRI recovery remains weak, identifying an important limitation of the learned classification (Sec. 5.2).

The inspection score S_{RG} combines radio-galaxy-like probability mass with ensemble reliability, distinguishing broad morphological support from concentrated and consistent predictions. Classes, score components and cutouts should be inspected together.

Among ten selected, masked relics, none reaches the high-score boundary. Their varied morphology assignments illustrate the model's response to deliberately isolated diffuse emission.

7.2 Future work

Independent review across classes and score bands should assess the quality of score-based selections and probability calibration against observed correctness (Guo et al., 2017). It should include sources outside the curated sample.

PyBDSF catalogue positions and fitted sizes offer a possible route to applying the pipeline without host identifications. Baron Perez et al. (2026, Sec. 2) centre DR3 cutouts on radio mean positions and use a side length of $2.5 \times \mathrm{Maj}$, where Maj is the fitted Gaussian major axis. Comparing these radio-centred inputs with our host-centred cutouts would test the sensitivity to source association, centring and size estimates.

Controlled retraining should compare intensity scaling and automatic source isolation on a fixed, independently reviewed sample. LAS-stratified performance would inform size-specific modelling, accounting for the smaller samples at large angular sizes.

Synthetic core-and-lobe images could also test sensitivity to central brightness, bending and faint extended emission, following the pre-training evaluation of Baron Perez et al. (2025). Any resulting training changes should be assessed on real sources, particularly reference FRI examples.

A broader relic sample with controlled masking and clipping variants would test how input preparation affects the predictions. Including unmasked fields, radio haloes and revived fossil plasma would extend the assessment to more varied diffuse-emission environments (van Weeren et al., 2019).

7.3 Data and software availability

The final source catalogue, trained classification model and author-developed analysis tools, including the polygon-masking interface, are available upon request; the inherited learning framework is credited separately in Chapter 1. Requests can be directed to `harutyunianedgar@gmail.com`.

Appendix A. GreyResNet18 encoder and projection head

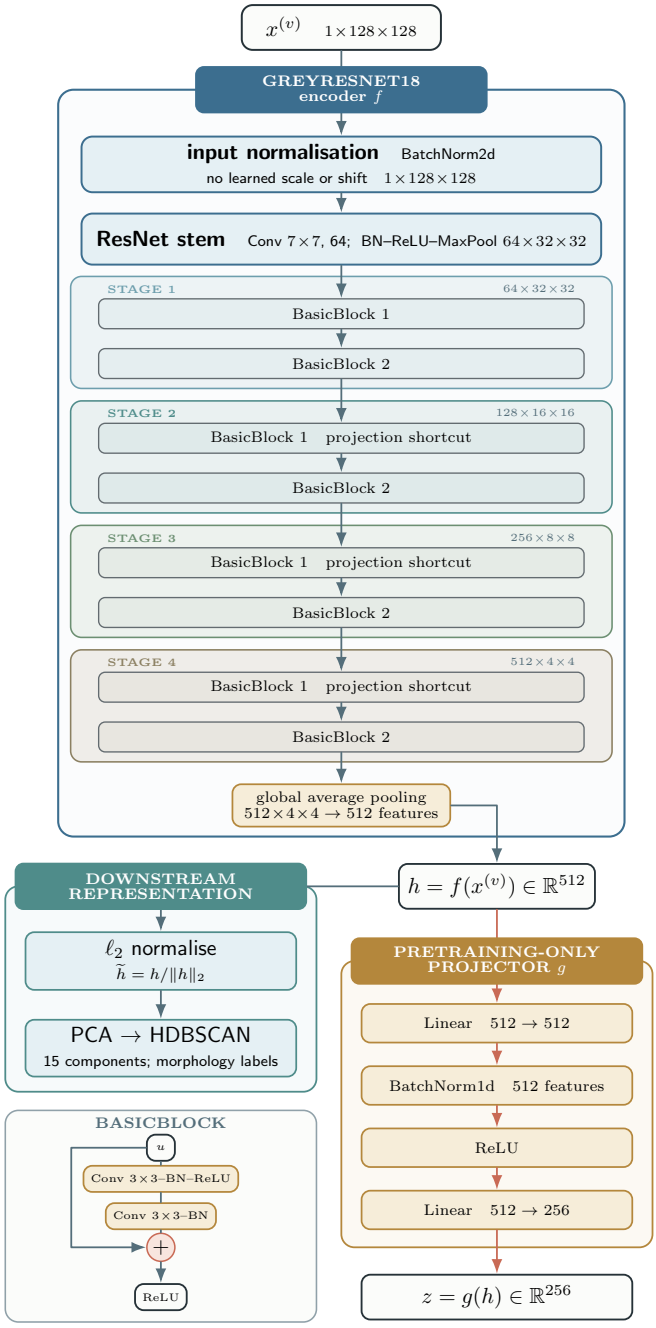


Figure 25: GreyResNet18 encoder and projection head used for contrastive pre-training, shown for one generic augmented view $x^{(v)}$; both SimCLR views pass through the same encoder and projector parameters. In the ResNet stem, the 7×7 convolution and max-pooling operation both use stride 2. The first BasicBlock of stages 2–4 halves the spatial resolution: its first 3×3 convolution and its 1×1 projection shortcut use stride 2, whereas the remaining BasicBlocks use stride 1 and identity shortcuts. Global average pooling is implemented by adaptive average pooling followed by flattening, producing the 512-dimensional representation h. The projection head g is used only during pre-training and maps h to the 256-dimensional projection z; it is discarded for downstream analysis.

Appendix B. Raw PCA–HDBSCAN cluster overview

Figures 26 and 27 show the raw groups used during manual inspection, with actual HDBSCAN labels $-1, 0, \ldots, 58$. Table 16 records their consolidation into the eleven morphology classes.

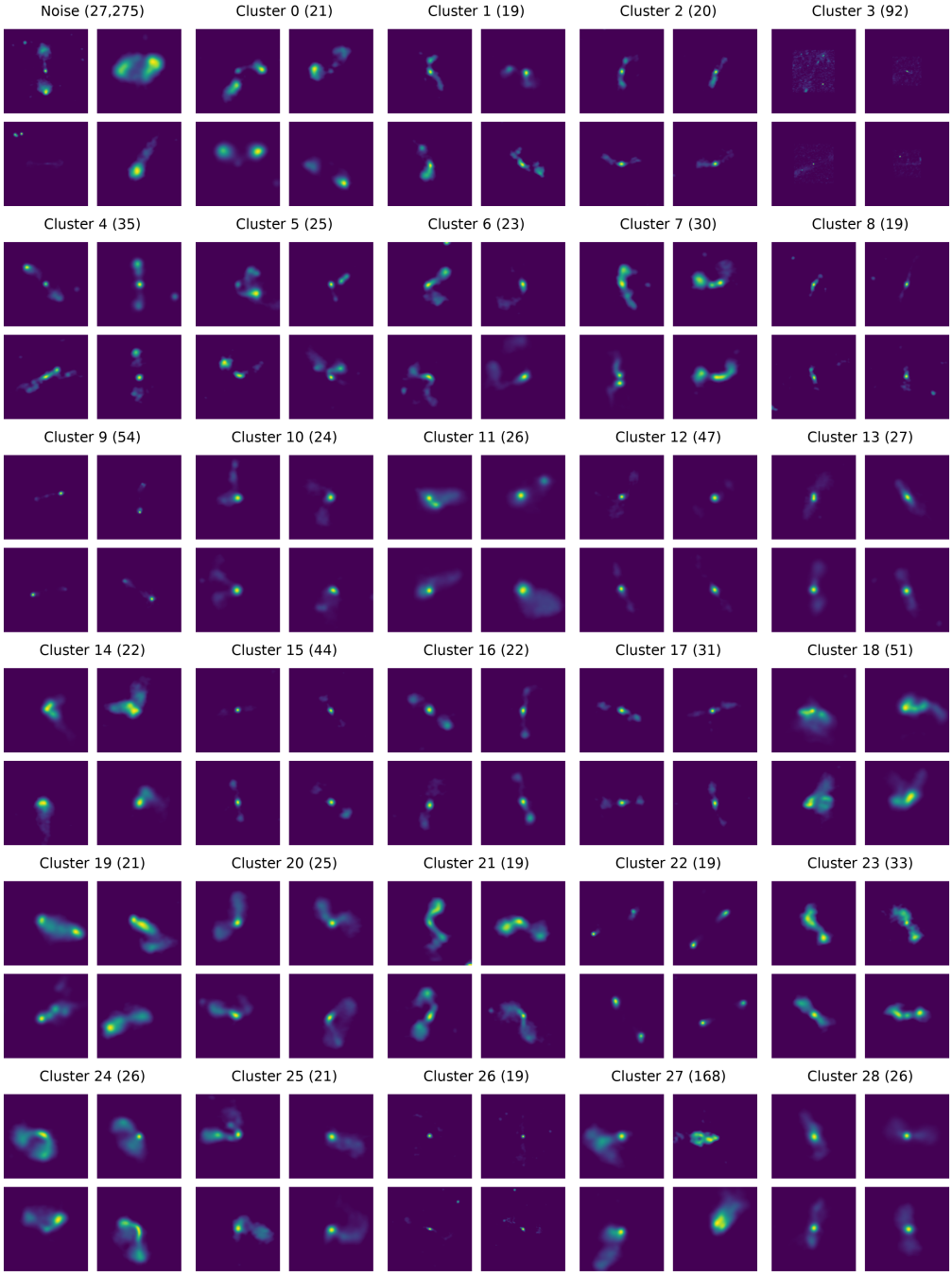


Figure 26: Noise group -1 and clusters 0–28, read left to right and top to bottom. Each group shows four examples; parentheses give its total membership.

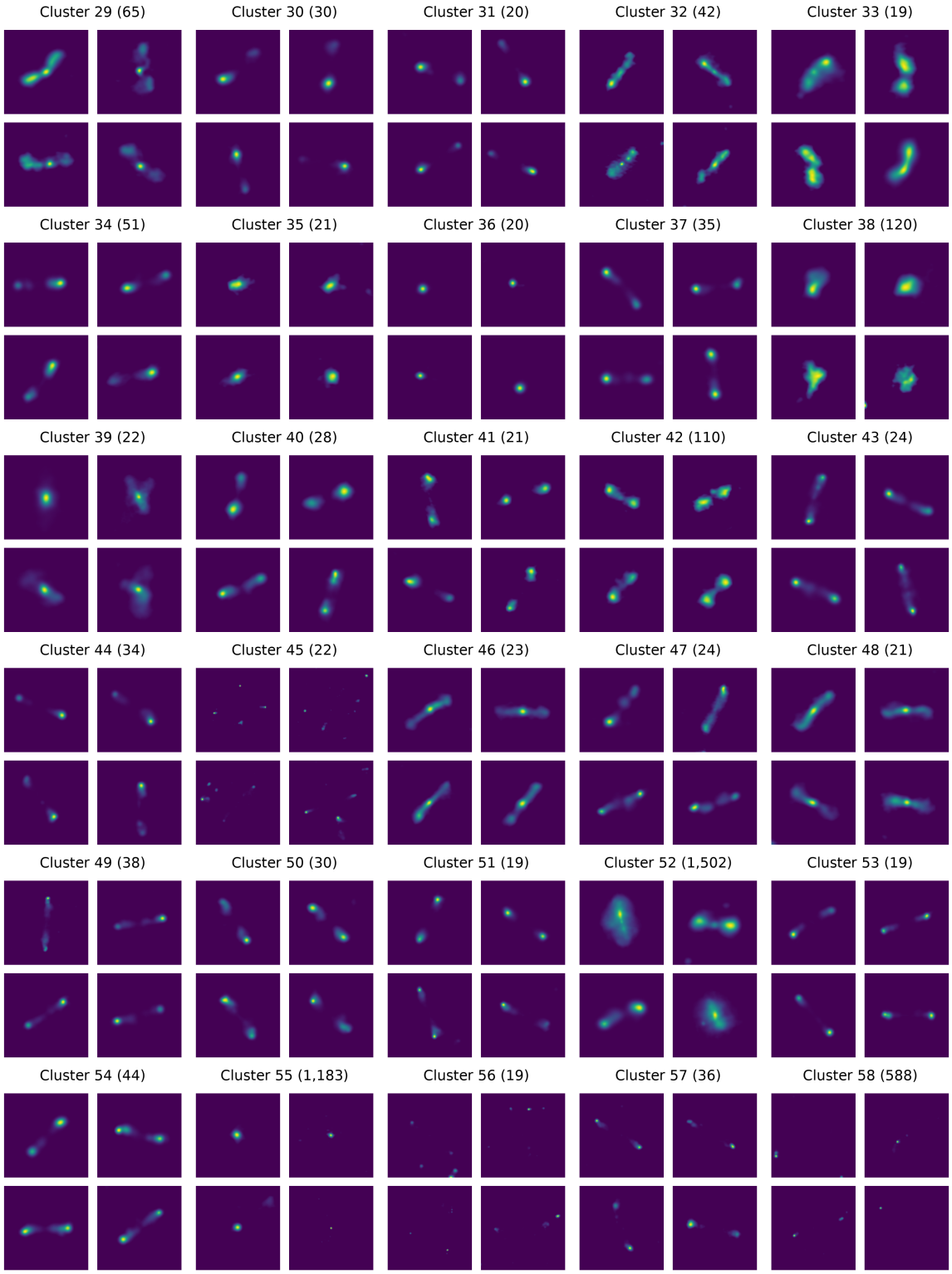


Figure 27: Clusters 29–58, in the same format as Figure 26.

Morphology class	Raw HDBSCAN labels	Sources
Amorphous	35	21
ACJ	11, 14	48
Bright-core	55	1 183
Centre-bright FRI	2, 4, 8, 12, 13, 15, 16, 17, 26, 28, 46, 48	334
Core-Displaced	36	20
Edge-bright FRII	22, 30, 31, 34, 37, 40, 41, 42, 43, 44, 47, 49, 50, 51, 53, 54	546
Hybrid	23, 32	75
Large FRII	9, 57	90
Spirals	38	120
Structured-Bent	0, 1, 5, 6, 7, 10, 20, 21, 24, 25, 29	298
Unresolved	3, 45, 56, 58	721
Returned to unassigned	18, 19, 27, 33, 39, 52	1 783
Original noise	-1	27 275

Table 16: Raw-cluster consolidation before SI-SSIM filtering. Every raw label occurs once; source counts sum to $32\,514$.

Appendix C. Examples of external-label disagreement

Figures 28–30 show five examples per internal class with at least one mapped-label disagreement. Selection favours reference coverage and label diversity. Row labels give ensemble predictions; Cle, Hor, and Chi denote Clews, Horton, and Chilufya, with a dash for missing mapped labels. C denotes confidence. The two internal FRII classes remain separate in this inspection sample.

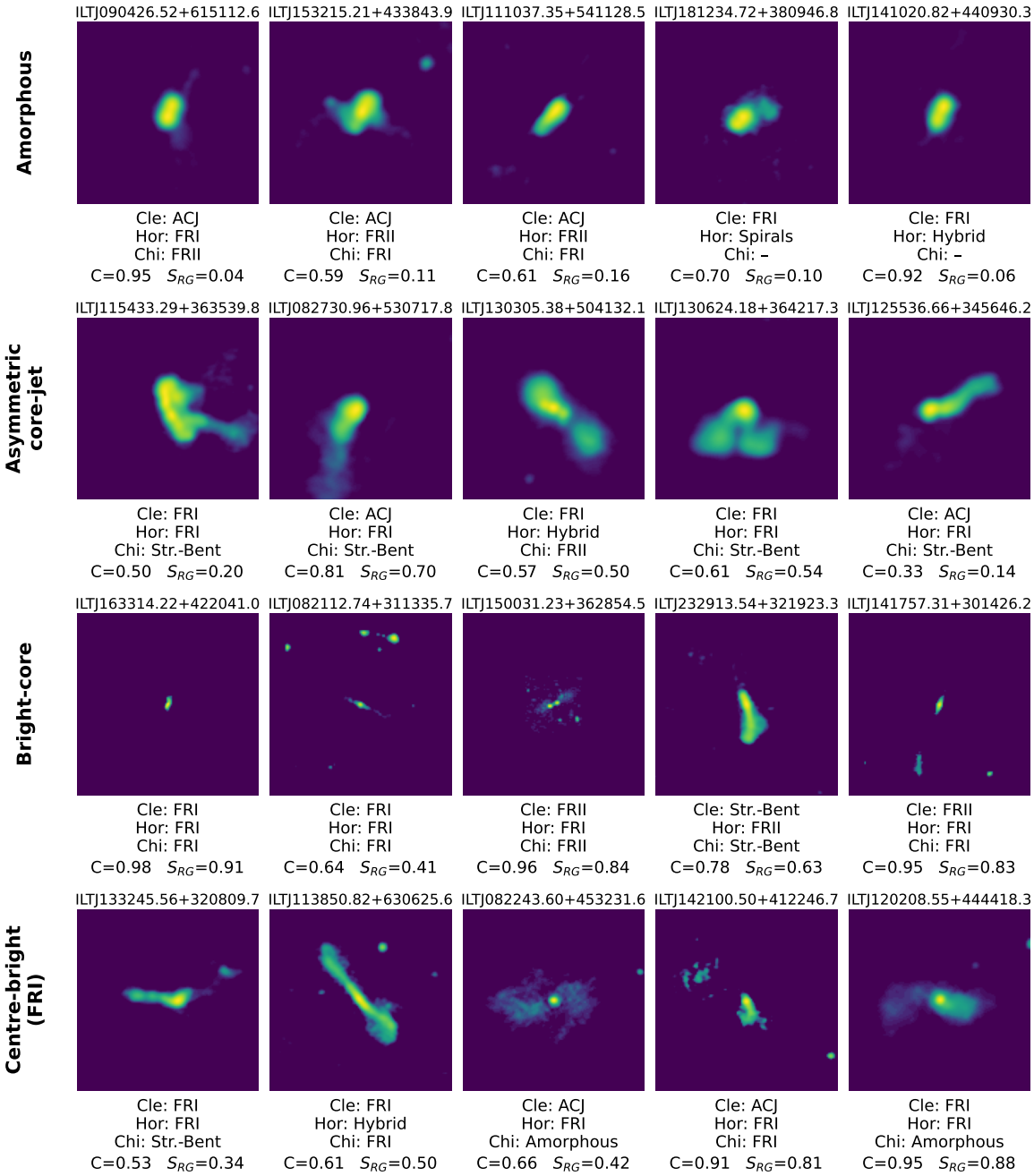


Figure 28: Five disagreement examples per class for Amorphous, ACJ, Bright-core, and FRI.

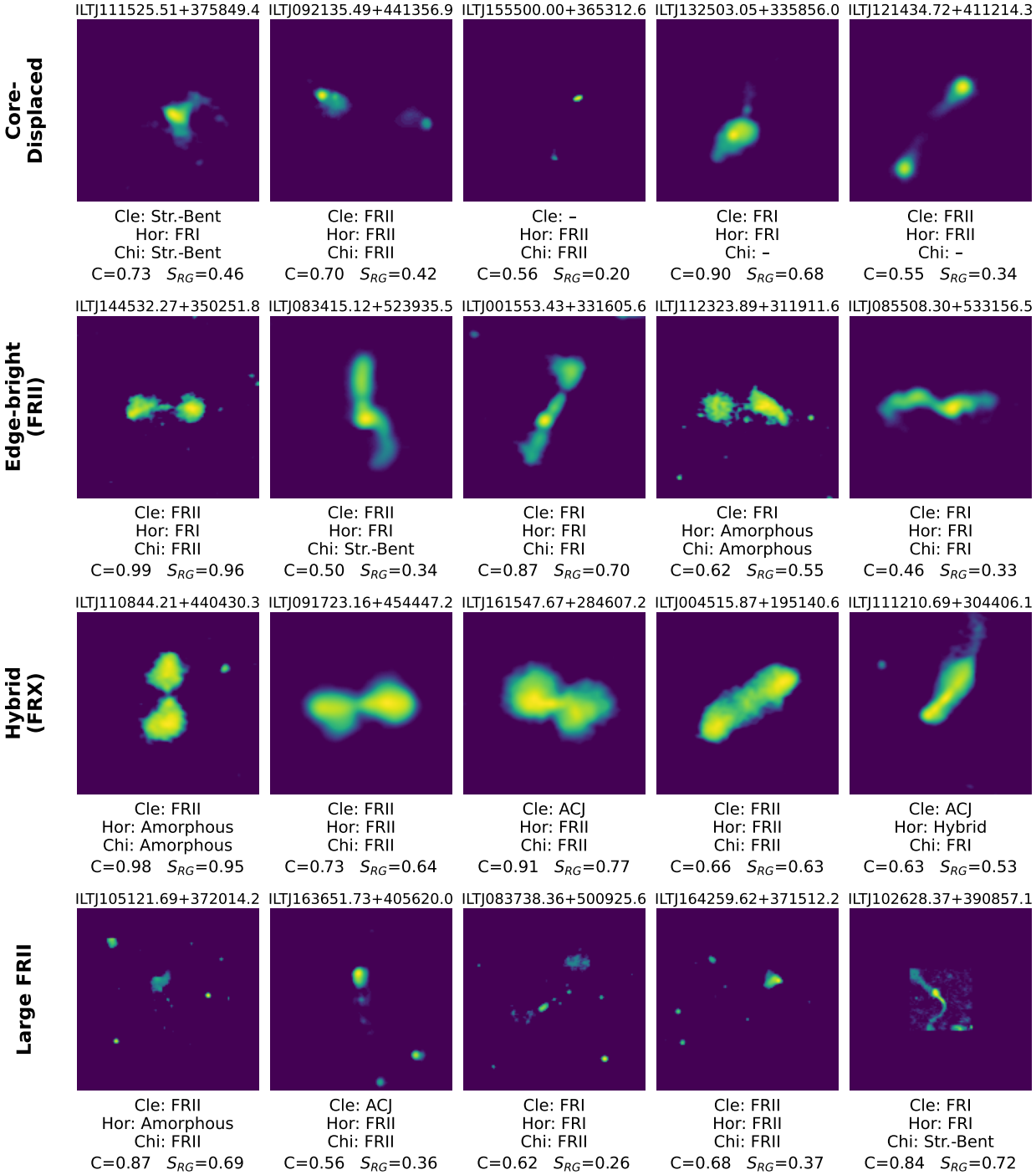


Figure 29: Disagreement examples for Core-Displaced, FRII, Hybrid, and Large FRII. Annotations follow Figure 28.

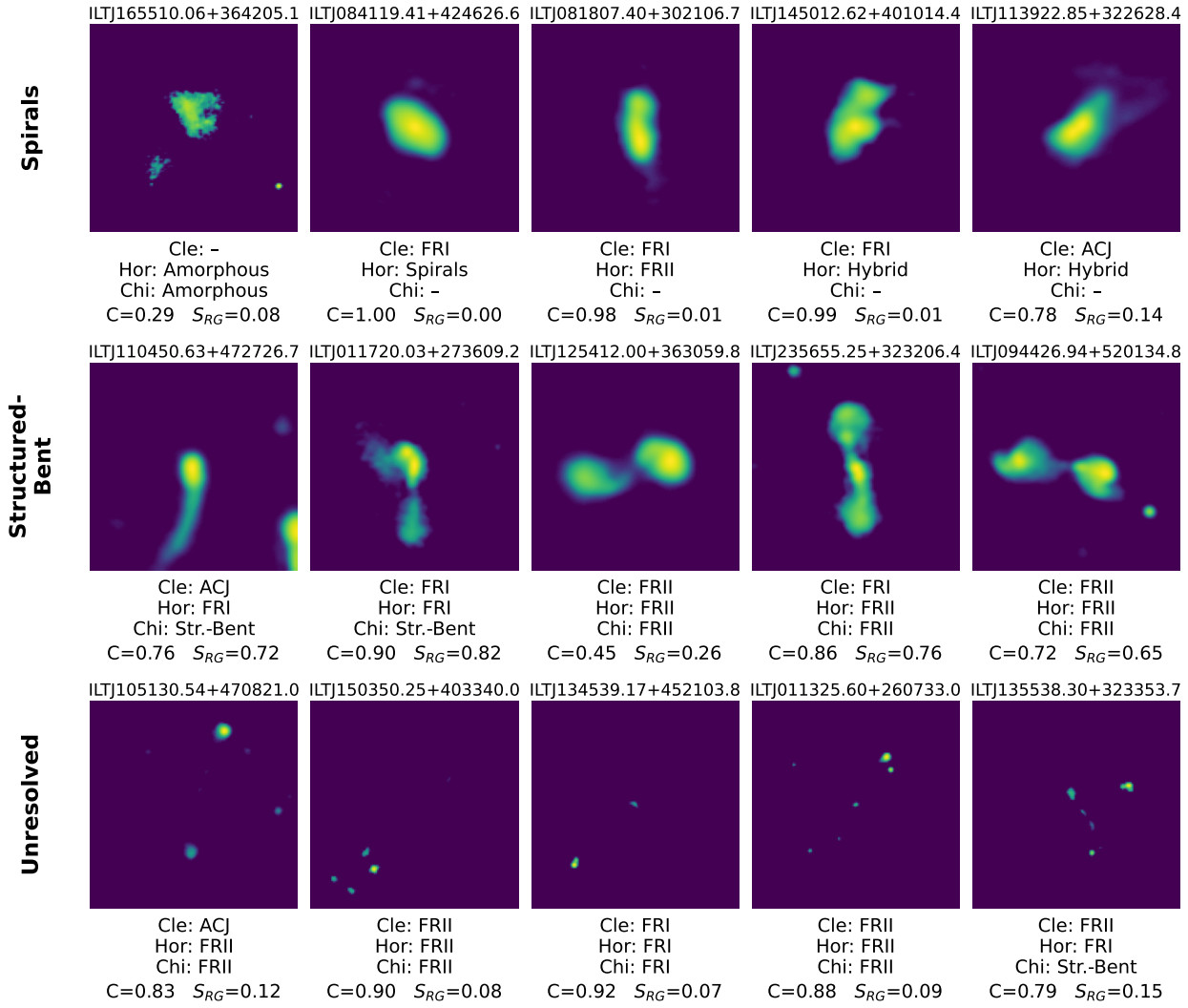


Figure 30: Disagreement examples for Spirals, Structured-Bent, and Unresolved. Annotations follow Figure 28.

Appendix D. Supplementary evaluation

The test-set counts in Table 17 use the same ensemble as the main catalogue. Each reference class contains three test sources; the source-level predictions and full confusion matrix are retained with the accompanying analysis outputs.

Reference class	Test sources	Correct	Other assignments
Amorphous	3	3	–
ACJ	3	1	Spirals (1), Structured-Bent (1)
Bright-core	3	3	–
FRI	3	2	Hybrid (1)
Core-Displaced	3	3	–
FRII	3	3	–
Hybrid	3	3	–
Large FRII	3	3	–
Spirals	3	3	–
Structured-Bent	3	1	ACJ (2)
Unresolved	3	3	–

Table 17: Predictions for the 33 test sources.

The F_1 uncertainties in the main comparison tables and Table 18 use source-level bootstrap resampling: for each catalogue separately, we repeatedly draw samples of the original size, allowing a source to be selected more than once. Each selected source retains its reference label and saved model prediction, and we recalculate F_1 for each sample without retraining the model. The central $95\,\%$ of the resulting F_1 values defines the reported uncertainty interval.

Signed uncertainties give the distances from each estimate to its lower and upper $95\,\%$ interval bounds. They describe sampling variation within the selected samples, excluding uncertainty in reference labels, source selection and training. FRII includes both internal FRII classes, as in the main comparisons.

Catalogue	LAS (arcsec)	Sources	FRI F_1	FRII F_1
Clews	60–120	6002	$0.310^{+0.024}_{-0.024}$	$0.739^{+0.013}_{-0.014}$
Horton	60–120	4635	$0.120^{+0.026}_{-0.025}$	$0.772^{+0.011}_{-0.012}$
Chilufya	60–120	1420	$0.388^{+0.057}_{-0.059}$	$0.378^{+0.044}_{-0.046}$
Clews	120–240	1539	$0.562^{+0.035}_{-0.036}$	$0.739^{+0.029}_{-0.032}$
Horton	120–240	2868	$0.383^{+0.033}_{-0.032}$	$0.719^{+0.018}_{-0.019}$
Chilufya	120–240	906	$0.490^{+0.050}_{-0.051}$	$0.521^{+0.051}_{-0.055}$
Clews	240–1000	216	$0.419^{+0.093}_{-0.105}$	$0.583^{+0.092}_{-0.099}$
Horton	240–1000	849	$0.363^{+0.054}_{-0.055}$	$0.605^{+0.042}_{-0.045}$
Chilufya	240–1000	264	$0.439^{+0.093}_{-0.101}$	$0.592^{+0.081}_{-0.089}$

Table 18: Curated-excluded comparisons by LAS, with $95\,\%$ bootstrap intervals. Bins include their lower edge; the final bin includes $1000''$.

Removing Clews's uncertain FR labels as well as all curated sources preserves the stronger FRII correspondence. This checks sensitivity to reference selection without resolving the remaining mapping ambiguities.

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